



केंद्रीय भूमि जल बोर्ड

**जल संसाधन, नदी विकास और गंगा संरक्षण
विभाग, जल शक्ति मंत्रालय**

भारत सरकार

Central Ground Water Board

**Department of Water Resources, River
Development and Ganga Rejuvenation,
Ministry of Jal Shakti
Government of India**

AQUIFER MAPPING AND MANAGEMENT OF GROUND WATER RESOURCES

EAST SINGHBHUM,

**SARAIKELA-KHARSAWAN AND PARTS OF WEST-
SINGHBHUM DISTRICTS, JHARKHAND**

राज्य एकक कार्यालय, रांची

State Unit Office, Ranchi

1.0 INTRODUCTION

With the growth of the multi-sectoral use of ground water, the urgency of adopting strategies to manage it in a sustainable way has become the necessity in the present scenario. It was felt mandatory to know the mode of occurrence, movement, quality and various aspects of ground water in East Singhbhum, Saraikela-Kharaswan and Parts of West-Singhbhum district. With this thought in the backdrop, a multi-disciplinary approach has been adopted in the form of 'National Aquifer Mapping' under which the studies have brought out the existence of potential ground water zones in the hard rock formations, its aerial extent, aquifer disposition with aquifer potentiality, ground water flow modeling studies in the multi aquifer system has brought an optimal ground water development management plan through exhaustive database generation as part of this study.

NAQIM Area Phase III

Legend

- Location Block HQ
- Railway
- Hiway
- District Boundary
- OE Block
- Block Boundary

SARAIKEELA KHARSAWAN

Parts of West Singhbhum

EAST SINGHBHUM

0 5 10 20 30 40 Km

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1.2 AREA DETAILS & DEMOGRAPHY (Location, Extent and Accessibility)

A total area of 7905 Sq.km was taken up under NAQUIM studies (Phase III) in East – Singhbhum district (3562 Sq.km), Saraikela-Kharsawan district (2659 Sq.km) and West-Singhbhum district (1684 Sq.km). The administrative set up of Phase III area is given in *Table 1*.

Table – 1 Administrative set up of Phase III area

S.No.	District Name	Geographical area (Sq. Km)	Village	No. of Tehsil	Name of Tehsil	No. of Blocks
1.	East-Singhbhum	3562	1765	11	Potamda, Boram, Golmuri-cum-Jugsalai, Ghatsila, Potka, Musabani, Dumaria, Dhalbhumgarh, Gurbandha, Chakulia, Baharagora,	11
2.	Saraikela-Kharsawan	2659	1191	9	Kuchai, Kharsawan, Chandil, Ichagarh, Kukru, Nimdih, Gamharia, Saraikela, Rajnagar	9
3.	West Singhbhum (Parts)	1684	666	5	Khuntpani, Chaibasa, Tantanagar, Bandgaon, Chakardharpur	5

The census report 2011 states that total population of the study area is **38,20,816** with rural population of 22,87,478 and urban population of 15,33,337. Based on the census -2011 district wise population details are given in *Table 2*.

Table-2 District- wise Population Details

Sl. N o.	District	Male	Female	Rural	Urban	Total
1.	East-Singhbhum	11,76,902	11,17,017	10,19,328	12,74,591	22,93,919
2	Saraikela-Kharsawan	5,44,411	5,20,645	8,06,310	2,58,746	10,65,056
3.	West Singhbhum	228536	233304	461840	-	461840
Total		19,49,849	18,70,966	22,87,478	15,33,337	38,20,816

1.3 DATA AVAILABILITY, DATA GAP ANALYSIS & DATA GENERATION

Data availability, data gap analysis and data generation and targets as per EFC norms and actual achievements are given below in *Table 3*. There is no aquifer-wise data available from State or any other agencies.

Table-3 Data availability, data gap analysis and data generation

Number of Blocks	25			
	Required	Existing	Balance	Actual achievements
Area in sq.km	7905			
Preparation of Sub-surface Geology (area in sq.km)	7905	0	7905	
Geo-morphological analysis (area in sq.km)	7905	0	Layer received	
Land use pattern (area in sq.km)	7905		Layer received	
Vertical electrical Sounding (VES) (Nos)	100	50	50	43
Bore Hole Logging (Nos)		Need based	Need based	
2-D Imaging (Line Km)				
Ground TEM (Nos)				
Heliborne TEM & Gravity (Line km)				
Preparation of Drainage Map (area in Sq.km)	7905		Layer received	
Demarcation of water bodies (area in sq.km)	7905		-	
Soil infiltration studies (Nos)	10		-	
Rainfall data analysis for estimation of recharge to ground water (area in sq.km)			-	
Canal flow, impact of recharge structures analysis, surface water-ground water interaction studies etc (area in sq.km)			-	
Water level monitoring (No of stations* frequency) (No of monitoring stations)	200@100*2		200@100*2	
Exploratory Wells	57	28	29	20
Observation Wells	29	9	20	7
Slug test (Nos)				
Specific Yield test (Nos)				
Micro-level hydro-geological data Acquisition including Quality Monitoring	100		100	
Water Quality (sampling and Analysis) for Basic Constituent, Heavy Metals etc	79	22	57	57
Analysis of Ground water for Pesticides, Bacteriological contamination in Ground water (Nos)				
Carbon dating (Nos)				
Isotopic studies (Nos)				
Core drilling (Nos)		Need based		

1.4 RAINFALL

Average annual Rainfall of the study area varies from 1200-1250 mm. Normal rainfall of East-Singhbhum district is 1312 mm. Normal rainfall of Saraikela-Kharaswan district is 1163 mm. Details of rain fall data of last five years for East-Singhbhum ,Saraikela-Kharaswan and West-Singhbhum districts is given in Table 4, 5 and 6 respectively.

Table -4 Rain fall data of East-Singhbhum district (2010-2014) (mm)

YEAR	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sep	Oct.	Nov.	Dec.
2010	0.5	6.2	5.8	10.4	74.1	51.7	99.6	133.1	137.2	48.3	10.6	38.6
2011	14.3	5.7	12.2	29.0	113.7	466	194	478	422	75	0.0	0.0
2012	42.4	17.6	0.0	50.4	34.8	189	418	291	268	40.6	50	22
2013	0.0	12.4	1.8	37.4	184.4	229	334	372	327	349	0.0	0.0
2014	0.9	75.2	23.6	0.5	116.2	231	318	281	126	166	0.0	0.0

Table-5 Rain fall data of Saraikela-Kharsawan district (2010-2014) (mm)

YEAR	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sep	Oct.	Nov.	Dec.
2010	0.0	0.0	0.0	0.0	0.0	0.0	28.4	88.5	133.3	175.5	37.5	0.0
2011	0.0	0.0	0.0	0.0	0.0	0.0	357.0	237	307	0.0	0.0	0.0
2012	0.0	31.0	0.0	0.0	0.0	0.0	336.3	269	224	8.5	13	24.2
2013	0.0	0.0	0.0	0.0	0.0	143.6	179.4	280	261	251	0.0	0.0
2014	0.0	8.2	16.1	0.0	46.5	165.6	321.3	291	105	66	0.0	0.0

Table-6 Rain fall data of West-Singhbhum district (2010-2014) (mm)

YEAR	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sep	Oct.	Nov.	Dec.
2010	0.0	2.2	3.0	36.4	0.0	56.3	100	180	172	55.6	0.0	42.0
2011	3.1	0.0	29.9	65.5	71.9	268	229	313	363	0.0	0.0	0.0
2012	38.6	7.4	0.0	8.6	0.0	146	165	256	200	27.2	48.9	29.5
2013	5.5	5.3	0.6	45.3	78.6	244	240	167	233	383.2	0.0	0.0
2014	0.0	54.4	26.4	2.1	80.3	282	411	457	160	46.0	0.0	0.0

(Source IMD)

1.5 PHYSIOGRAPHY

Most of the hills are residual. The northern Dalma ridge is the dominant volcanic hill form a mountain range of considerable prominence above the Tertiary plain. WNW-ESE trending Dalma hill range is the most prominent hill range in the area. A significant part of the range were above the 600 m elevation. Generally the elevation ranges from 213 m to 945 m above mean sea level. About 53% of the total area of the district is covered by residual hills consisting of granite, gneiss, schist and basalt rocks.

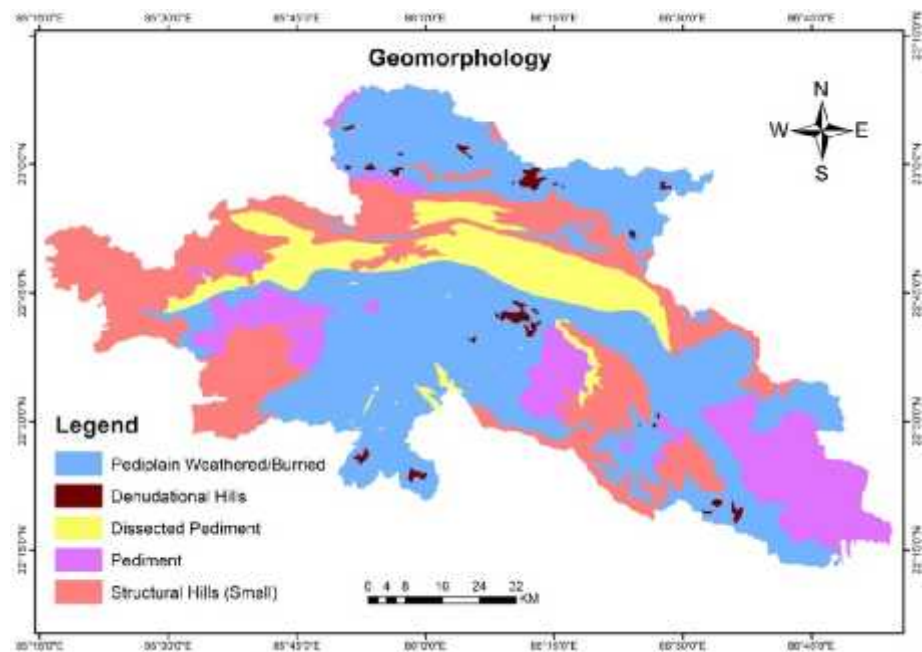


Figure 2 Geomorphological map

1.6 DRAINAGE

The Subarnarekha river flows from west to south-east direction. All the tributaries of this area meet with the Subarnarekha river. Drainage pattern is dendritic in nature. Drainage of Potamda block do not meet in Subarnarekha River. Kharkai River meets Subarnarekha river at Sonari near Jamshedpur. Major tributaries which meet Subarnarekha river from west to east are Sapnara nadi, Garra nadi, Dudh nadi, Chakdaha nadi. The principal rivers of the district are Subarnarekha and Kharkhai Rivers. The general trend of the drainage is from NW-SE and SW-SE. The structural features particularly the foliation and joints exert profound impact upon the drainage and control the drainage pattern of the district. The Drainage map is shown in figure no 3



Figure 3 Drainage map

1.7 LAND USE

The total geographical area of the study area (NAQUIM phase- III) is 7905 sq km. Gross cropped area is 1651 sqkm whereas net sown area is 1361 Sq.km. Details of land use pattern is given below in *table 7* and land use map is shown in *figure no 4*

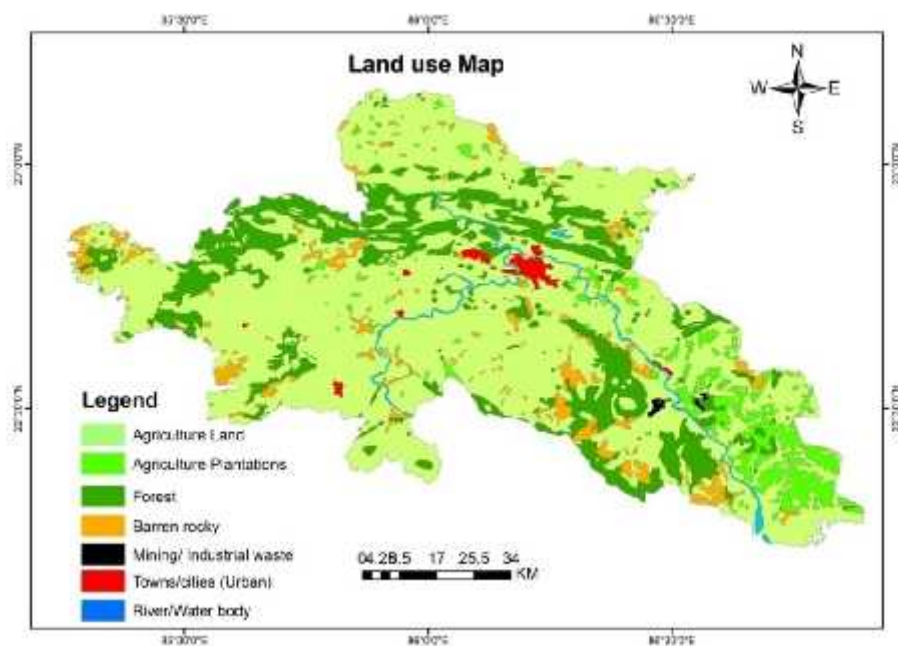


Figure 4 Land use map

Table 7 – Details of Land use

S. no	District	Geographical area (ha)	Net sown area (Ha)	Gross crop area (ha)	Cropping intensity %	Gross Irrigated area (ha)	Forest area (ha)
1	East Singhbhum	356200	85632	98057	114	9784	1123
2	Saraikela	265900	50470	67086	132	5415	654
3.	Parts of W.Singhbhum	168400	13584	14209	104	1594	495

1.8 SOIL

Five types of soils are found in the study area, 1. Red gravelly, 2. Red sandy, 3. Red loamy, 4. Red and Yellow 5. Lateritic soil.

Red gravelly soil is found in Chakulia block and parts of Bahragora blocks. Red sandy soil is observed in Mosabani, Parts of Jamshedpur and Dumaria blocks. Red loamy soil is found in parts of Bahragora, Dhalbhumgarh and Jamshedpur sadar blocks. Red and Yellow soil is found in Patamda and Potka blocks. Lateritic soil is found in small patch of Bahragora block.

1.9 IRRIGATION

The study area of phase III has poorly developed irrigation facilities. According to the fourth minor irrigation census data total 31142 ha is irrigated by minor irrigation schemes whereas 6576 ha is irrigated by ground water schemes and 24566 ha is irrigated through surface water schemes. The details of area irrigated by different schemes and total numbers of sources is given in (Table 8 & 9)

Table 8 Details of area irrigated by different schemes

District	Geographical area (ha)	Area irrigated by G.W. scheme in (ha)	Area irrigated by S.W. scheme in (ha)
East Singhbhum	356200	3147	6637
Saraikela	265900	1448	3967
Parts of West Singhbhum	168400	1981	13962

Table 9 Details of total numbers of Irrigation sources

District	Dugwell Shallow	S. Tubewell	Deep Tubewell	Total	S.Flow Scheme	S.Lift Scheme	Total	Grand Total
E.Singhbhum	1019	541	76	1636	1634	488	2122	3758
Saraikela	1674	27	2	1703	2088	149	2237	3940
Parts of W.Singhbhum	405	3	0	408	1284	-	1284	1692

1.10 CROPPING PATTERN

In the study area of Phase III agriculture and mining is the main occupation of the people. The cropping pattern is generally influenced by varied soil type and different climatic conditions. The principal crop of the area are Paddy, maize, Pulses and Oilseeds. The horticulture crops are Cauliflower, cabbage, tomato, Brinjal, L. finger, cucumber etc. The details cropping pattern is given in *Table 10*.

Table 10 Details of cropping pattern

District	Geographical Area (ha)	Net sown area (ha)	Season	Crop	Cropped area (ha)	% of cropped
E. Singhbhum	356200	85632	Kharif	Paddy	22974	27
				Maize		
				Pulses		
			Rabi	Wheat	3290	4
				Pulses		
				Oilseeds		
				Pulses		
Saraikela	265900	50470	Kharif	Paddy	8514	17
				Maize		
				Pulses		
			Rabi	Wheat	2760	5
				Pulses		
				Oilseeds		
W. Singhbhum	168400	13584	Kharif	Paddy	4022	29
				Maize		
				Pulses		
			Rabi	Wheat	2457	18
				Pulses		
				Oilseeds		

2.0 DATA COLLECTION & GENERATION

2.1 GEOLOGY

The Singhbhum Granite Complex is surrounded on the east, north and north-east predominantly by pelitic supra-crustals with granitic intrusive designated as Proterozoic Mobile Belt or North Singhbhum Mobile Belt (NSMB) extending roughly in east –west direction and delimited by Singhbhum Shear Zone (SSZ) in the south. Chhotanagpur Gneissic Complex (CGC) in the north which is delimited by Tamar Porapahar Shear Zone (TPSZ). The formation north of Archean Cratonic margin (Singhbhum Granitic Complex) and south of Chhotanagpur Granite Gneissic Complex are now referred as North Singhbhum Mobile Belt (NSMB) which evolved during the Proterozoic which has five litho-stratigraphic domains which are: Belt of volcano-sedimentary formations (IOG and Dhanjori rocks) wrapping Archean cratonic mass to the north bounded by Singhbhum Shear Zone. Rocks of Singhbhum Shear Zone (SSZ), Rocks of Chaibhasa and Dhalbhum Formations between Singhbhum Shear Zone (SSZ) and Dalma volcanic. The entire Dalma volcano–sedimentary belt comprising low grade metamorphosed basic and ultrabasics volcanics. Low to medium grade volcano-sedimentary belt between Dalma and Chhotanagpur Gneissic Complex (CGC). The main rock types found in the Singhbhum shear zone are quartz-chlorite-schist, banded quartzite, quartzite-talc-chlorite-Schist, quartz-sericitechlorite-schist with varying amount of magnetite and apatite. The quartzites are folded and shows bedding planes and groove lineation. The evidence of shearing is manifested by presence of crushed rocks and intense silicifications. The Iron Ore Group comprises an assemblage of Archean supracrustal comprising mafic-ultramafic volcanics, volcanoclastics, chert, BIF, greywackes, meta-argillites and arenites. The metasedimentary rocks within this Proterozoic Mobile belt (NSMB) can be divided into the Chaibhasa Formation and Dhalbhum Formation. The geological map of study area of phase III is shown in *figure 5*. A general stratigraphic sequence of this area is given below-

Generalized stratigraphic succession of Archean Proterozoic of Singhbhum Province (Saha et al., 1988)

Group/Formation		Lithology	Age
Arkasani Granophyre, Soda Granite Kolhan Group, Gangpur Group		Granophyre Sandstone, Shale, Limestone Quartzite, dolomite, phyllite, conglomerate	c.2.1 Ga
Dalma/ Dhanjori/ Ongarbira Group		Mafics, ultramafic, quartzites	
Singhbhum Group (Bonai Group?)		Dhalbhum Fm: Carbonaceous phyllite, quartzites, cherts, epidiorites, acid volcanics	
		Chaibasa Fm: Mica schist, quartzites, hornblende schist	c. 2.4-2.3 Ga
Newer Dolerite dykes and sills (Undifferentiated)			c.3.1 Ga
Iron Ore Supergroup	Iron Ore Group (Lower Bonai Group) Gorumahishani Group	Tuffaceous shale, phyllite, tuffs, BIF, Ferruginous chert, local dolomite, acid-intermediate, basic-ultrabasic volcanics, quartzites and conglomerate, quartzite, BIF, metamorphosed mafic and ultramafic volcanics	
Singhbhum Granite (SBG-A) (Phase I & II)/ Chakradharpur Granite		Granite/Granite gneiss	c.3.3 Ga
Older Metamorphic Tonalite Gneiss (OMTG)		Tonalite Gneiss	c.3.4-3.5 Ga
Older Metamorphic Group (OMG)		Pelitic schist, quartzites, paraand ortho-amphibolites	(folding and metamorphism of OMG and OMTG)

water occurs both under unconfined condition and semi confined to confined condition. The unconfined condition exists in the weathered mantle portion of the rocks. Depth of weathered mantle varies from 15-34 m in general. In Hard rock terrain fractures encountered within the depth 30-200 mbgl. The piezometric head or water level of shallow aquifer ranges between 3.3-20 mbgl and yield ranges 0.45 to 11.5 lps.

2.2 DEPTH TO WATER LEVEL:

Ground water regime is monitored through 39 dug wells in the study area. With the field data, maps were prepared for visual interpretation of the behavior of the ground water levels. Depths to ground water level were demarcated into various zones in the ranges of less than 2 m, 2-5 m, 5-10 m, 10-20 m. The description of depth to water levels during pre-monsoon and post monsoon is as follows:

2.2.a Depth to Water level May 2015:

Depth to ground water level during May 2015 ranges from 2 mbgl to 15.18 mbgl. Minimum depth to water level 2 mbgl recorded at Barma mine Thana in East Singhbhum district and maximum depth to water level 15.18 mbgl recorded at Chakulia East Singhbhum district. Depth to water level map May 2015 is shown in *figure 6*.

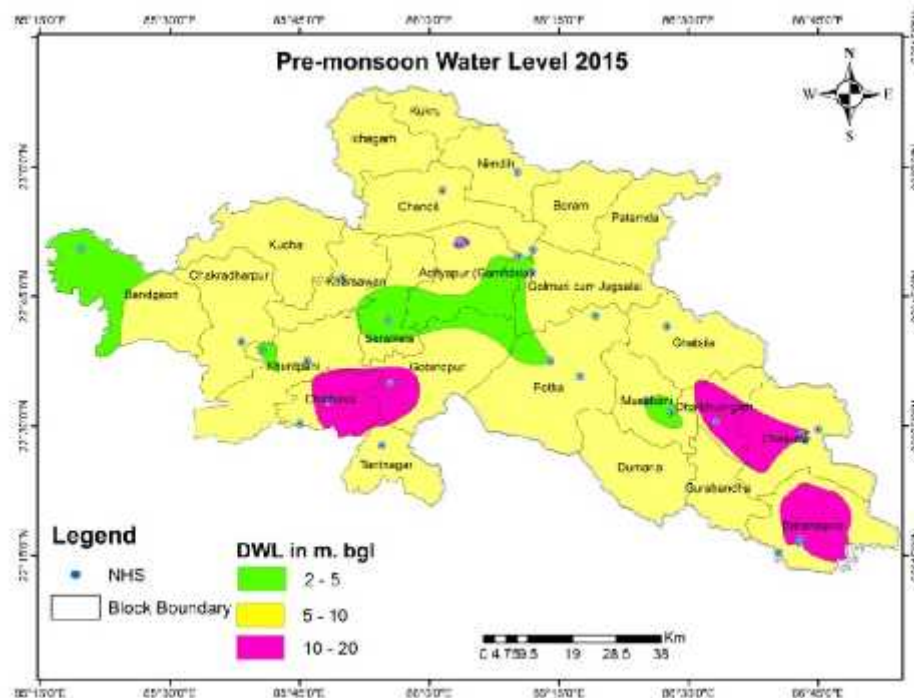


Figure 6 Depth to water level Map (Pre-monsoon-2015)

2.2.b Depth to Water level November 2015:

During month of November 2015 (post-monsoon) depth to water level varied from 1.49 m bgl to 17.1 m bgl. Minimum depth to water level 1.49 mbgl recorded at Barma mine Thana in East Singhbhum district and maximum depth to water level 17.1 mbgl recorded at Chakulia in East Singhbhum district. Depth to water level map November 2015 is shown in figure 7.

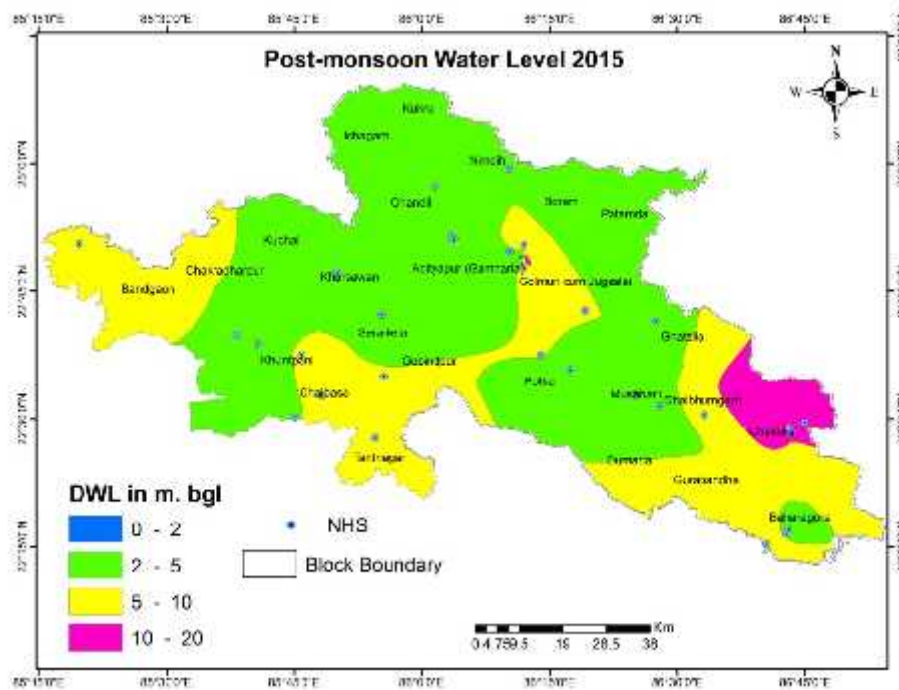


Figure 7 Depth to water level Map (Post-monsoon-2015)

2.3 HYDROLOGICAL TEST

2.3.1 Aquifer Performance Test (APT):

The aquifer performance test in wells tapping different litho-units have been carried out and the data analyzed using various methods as described by Jacob (time-drawdown), Theis (curve matching), depending upon time drawdown/recuperation behaviour. The results are given below in Table 11 and the result of the Aquifer performance Test in Hard Rock area is shown in figure 8.

Table 11 Result of the Aquifer performance Test in Hard Rock area

Litho-unit	Well	Depth of well (m)	S.W. L (mbgl)	Discharge (lps)	Draw down (m)	Specific Capacity (lps/m)	Transmissivity m^2/day	Storage coefficient	Pump Duration (min) & method adopted
Meta sediment	Kudada EW	145.4	2.42	8.83	18.21	0.48	19	6.91×10^{-4}	1620 (min) Thies
S. Granite	Hesel EW	204.7	2	3.68	45.84	0.08	4	-	1560 (min) Jacob
Ckp.Grani	CKP	210	6.9	1.2	34.46	0.034	2	-	720 (min)

te	EW								Jacob
S.Granite	Mahulbera EW	190.6	8.02	6.6	24.82	0.05	38.64	1.7×10^{-4}	200 (min) Jacob
S.Granite	Potka EW	198.7	4.89	5.2	9.5	0.04	26.48	-	500 (min) Jacob
S.Granite	Kandra EW	196.7	8.50	3	20.51	-	2.69	8.32×10^{-4}	120 (min) Jacob
S.Granite	Uliyan EW	190.7	11.50	4	19.47	-	15.5	-	200 (min) Jacob
Tertiary	Kala-pathar EW	100	4.05	12.2	12.30	-	207	-	400 (min) Jacob
Tertiary	Bahragora EW	99.25	3.63	21	14.44	-	570	-	-
Tertiary	Mansumaria Ew	121	12.34	15	10.98	-	209	1.30×10^{-4}	-

The tests indicate that the hydraulic behaviour of different hard rock aquifers i.e shallow and deep fractures are not identical. The salient features of the test results are as follows:

- i. No stabilization of pumping water level.
- ii. Generally steep draw down, but comparatively less in case of shallow fracture aquifers.
- iii. Differential rate of draw down, increasing magnitude with time. In case of shallow fracture aquifer showing uniform pattern of draw down upto initial 100 minute.
- iv. The rate of recuperation is very low, but comparatively faster in shallow fracture aquifers.
- v. There is variation in discharge during drilling and pumping test, the shallow fracture aquifer showing marginal increase of discharge whereas the deeper fracture aquifers exhibit dwindling discharge during the tests.

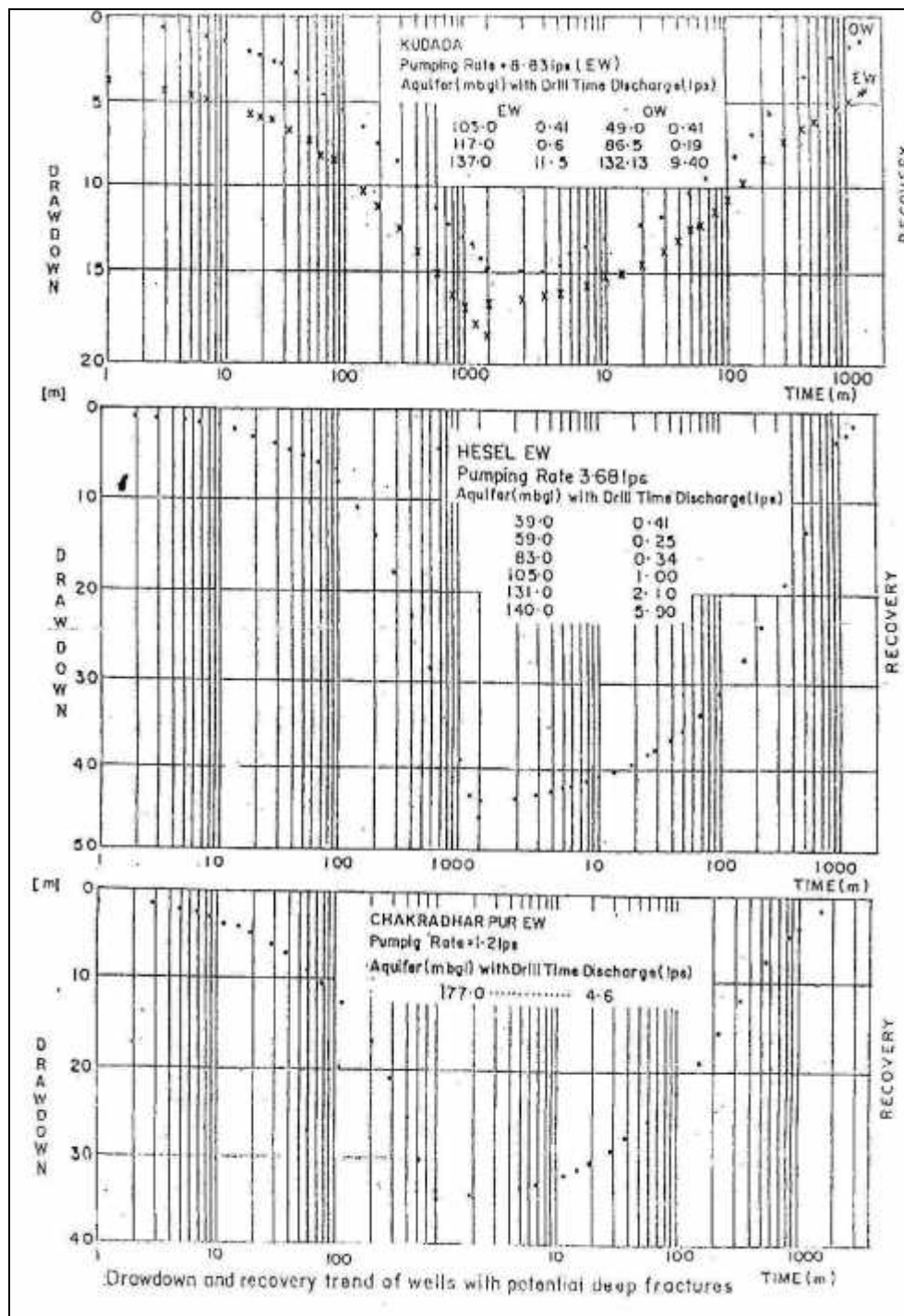


Figure 8 Result of the Aquifer performance Test in Hard Rock area

2.3.2 Step drawdown Test (SDT):

Step Drawdown test have been carried out in different wells drilled in hard rock terrain of study area of phase III. The result of the test are summarized in the Table 12

Table 12 Results of step Drawdown Test

Litho-unit	Location	Steps	Duraion (min.)	Discharge (lps)	Cumulative Drawdown (m)	Sp.capacity (lpm/m/)	Remarks
CKP Granite	CKP,EW	1	100	1.10	10.21	6.460	Aquifers deep fracture
		2	100	1.30	16.89	4.620	
		3	100	1.46	24.37	3.61	
		4	100	1.58	32.54	2.92	
	CKP,OW	1	100	5.75	6.38	54.07	Aquifers shallow, deep fracture
		2	100	6.66	9.32	42.89	
		3	100	7.58	11.97	38.01	
		4	100	8.50	14.51	35.15	
Singhbhum Granite	Hesel, EW	1	100	2.61	5.065	30.91	Aquifers deep fracture
		2	100	3.61	3.66	10.94	
		3	100	4.20	17.96	14.0	
		4	100	4.88	25.31	11.57	
Meta sediment	Kudada EW	1	120	4.68	4.98	56.385	Aquifers deep fracture
		2	120	6.87	8.076	51.046	
		3	120	8.80	10.48	50.381	
		4	120	11.0	13.49	48.925	

The result of the SDT indicates that there is vast variation in the specific capacity values in different formations.

2.4 HYDROCHEMICAL STUDIES

2.4.1 Water Quality Monitoring:

Water quality was being monitored through 57 monitoring wells which includes 22 hydrograph stations (dug wells) existing in the area. During the study additional 31 samples were collected for industrial cluster in Adityapur-Gamharia and Jamshedpur and 4 samples were collected through newly established key wells for analysis of general chemical parameters and trace elements. However, the data collected shows water quality of phreatic (weathered) aquifer only. There are few water quality data for fractured aquifer separately. An overview of Hydro-chemical data of (phreatic aquifer) hydrograph monitoring wells existing in Singhbhum Region is tabulated below in table no.13 and the details of water quality analysis is given in *Annexure I*. The spot value map of water quality samples more than permissible limit is also shown in *figure 9*. Hydro-chemical data of (phreatic aquifer) hydrograph monitoring wells existing in Singhbhum Region is tabulated below in *Table 13*

Table 13 Details of Hydro-chemical data of hydrograph monitoring wells

S.No.	Constituents	Minimum	Maximum	Average	BIS (2012)	
					Desirable	Permissible
1.	pH	7.12	8.66	8.11	6.5-8.5	No relax.
2.	EC	135	1866	808		
3.	TDS mg/l	87.75	1212.9	525.49	500	2000
4.	HCO ₃ mg/l	61.5	430	241	200	600
5.	Cl mg/l	7	350	95.5	98	1000
6.	TH (as CaCO ₃) mg/l	50	540	298	300	600
7.	Ca mg/l	16	150	74.6	75	200
8.	Mg mg/l	2.43	65.61	26.02	30	100

9.	Na mg/l	9.4	155	47.64	200	-
10.	K mg/l	0	35	9.03	200	-
11.	NO3 mg/l	1	66	18.9	45	No relax.
12.	F	0	1.32	0.402	1.0	1.5
13.	SO4	0	154	22.5	200	400

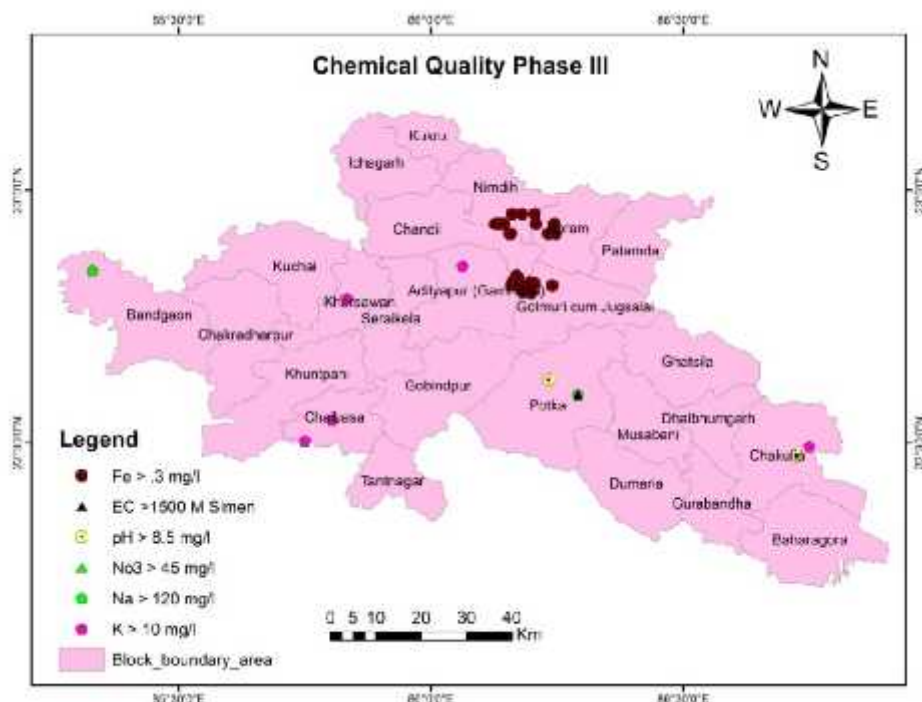


Figure 9 Chemical quality map showing more than permissible limit

Hydro-chemical data of (phreatic aquifer) Industrial Cluster samples collected in Singhbhum Region is tabulated below in Table 14.

Table 14 Details of Hydro-chemical data of Industrial Cluster samples of Adityapur-Gamharia & Jamshedpur Sadar blocks

S.No.	Constituents	Minimum	Maximum	Average	BIS (2012)	
					Desirable	Permissible
1.	pH	7.00	8.30	7.76	6.5-8.5	No relax.
2.	EC	302	2570	948.23		
3.	TDS mg/l	196.3	1670.5	616.35	500	2000
4.	HCO3 mg/l	123	522.75	259.3	200	600
5.	Cl mg/l	17.72	584.52	124.36	98	1000
6.	TH (as CaCO3) mg/l	105	2754	413.97	300	600
7.	Ca mg/l	18	188	67.27	75	200
8.	Mg mg/l	7.29	144.58	39.16	30	100
9.	Na mg/l	17.6	198	64.53	200	-
10.	K mg/l	.08	14.35	2.83	200	-
11.	NO3 mg/l	5.13	36.78	20.24	45	No relax.
12.	F	0.00	0.73	0.35	1.0	1.5
13.	SO4	13.25	111.53	56.27	200	400

Hydro-chemical data of (Shallow-Deeper aquifer) Bore-well samples collected in Singhbhum Region is tabulated below in Table 15

Table 15 Details of Hydro-chemical data of Exploratory Bore-well samples

Lithounit	Location	Aquifer	pH	EC	HCO ₃	Cl	TH	Ca	Mg	Na	K	NO ₃	F	So ₄
Meta sediment	Kudada, EW	FD	8.5	721	494	7	435	10	99	5	4	0.9	.6	5
Meta sediment	Kudada OW	FS,FD	8.6	607	311	7	370	6	86	9	3.7			
S.Granite	Hesel EW	FS,FD	8.6	251	159	12	130	32	12	28	25			
S.Granite	Hesel OW	FS,FD	7.8	185	85	14	75	20	6	8	.7			

FS : Shallow Fracture

FD: Deep Fracture

Hydrogen ion concentration (pH):

The pH of water indicates that whether the water is acidic or alkaline. The hydrogen (pH) is a measure of the hydrogen ion concentration in the water. The measurement scale of pH ranges from 1 to 14 with a pH of 7 indicating as neutral condition environment. The value of pH lower than 7 indicate acidic and more than 7 indicates alkaline. The pH of ground water in the study area ranged in between 7.00 to 8.66. The ground water of the study area can be assessed as slightly alkaline to neutral in nature.

Electrical Conductivity (Ec):

Electrical conductance is the ability of water to conduct electric current and it depends on, the concentration of ion, nature and types of ions and temperature. BIS has recommended desirable limit for Total Dissolve Solid (TDS) as 500 mg/l corresponding to EC value approximately as 750 μ S/cm at 25°C which is extendable to permissible limit as 2000mg/l. TDS corresponding to EC value is about 3000 μ S/cm at 25°C in absence of alternate source of water.

The electrical conductivity of ground water in Phase III area ranged in between 135 to 2570 μ S/cm at 25°C. Based on the electrical conductivity results the ground water in the study Area is potable.

Carbonate and Bicarbonate (Alkalinity):

Presence of alkalinity in water is the capacity to neutralize a strong acid due to the presence of carbonate, bicarbonate and hydroxide of magnesium and calcium. The carbonate concentration in ground water has not been deducted in the ground water and the concentration of bicarbonate ranged in between 61.5 to 522.75 mg/l.

Chloride:

Chloride in ground water can be geogenic in deep aquifers or caused by industrial or domestic wastes and pollution from brine. The BIS has set 250 mg/l chloride ions as acceptable limit and 1000 mg/l as permissible limit in the absence of alternate source in drinking water. In the study area Chloride concentration ranged in between 7 to 584 mg/l.

Fluoride:

Fluoride occurs in low concentration in natural water. It is an essential element for maintain normal development of teeth and bones. The consumption of low fluoride concentration prevents the dental caries. The higher concentration of fluoride consumption

causes mottling of teeth and skeletal fluorosis. The BIS has set the maximum concentration of fluoride in drinking water is 1.5 mg/l as permissible limit. The fluoride concentration in the study Area ranged in between 0.0 to 1.32 mg/l. The maximum concentration of fluoride has been recorded in Bharagora block of East Singhbhum district.

Nitrate:

Nitrate is the end product of oxidation process of nitrogen in the environment and its higher concentration in ground water is an indication of pollution from human and animal wastes, nitrogenous fertilizers and industrial wastes. Nitrate is a non-essential constituent of ground water. The BIS has set a acceptable limit of nitrate concentration in drinking water as 45 mg/l with no relaxation. The consumption of nitrate more than 45 mg/l is harmful for human being, particularly to infants as it may cause blue baby syndrome.

In the study area nitrate concentration in ground water ranges in between 1.00 to 66 mg/l. The maximum Nitrate concentration has been recorded at Sakchi in East Singhbhum district.

Sulphate :

The concentration of sulphate in drinking water set by BIS as acceptable limit is 250 mg/l and 400 mg/l permissible limit in absence of alternate source. Higher concentrations are undesirable taste because of laxative effects. Sulphates in groundwater are released by natural deposition of calcium sulphate, magnesium sulphate, or sodium sulphate. These natural sources can be applied as soil conditioners. The ground water of study area recorded sulphate concentration ranged in between 0.0 to 154 mg/l.

Total Hardness:

As per Bureau of Indian Standard (BIS): 10500-2012 (Drinking Water), the acceptable limit of total hardness is 300 mg/l and permissible limit is 600 mg/l in absence of alternate sources.

Total hardness of ground water in the study area ranges in between 50 to 2754 mg/l. The maximum concentration of total hardness above acceptable limit was observed at Gangpur village (Dw) opposite to jamujara lubricant private company, in Adityapur-Gamharia district.

Sodium:

Sodium does not find freely in nature because it is very active element. It is always combines with other elements. In human body, sodium helps in maintaining water balance. The higher sodium intake may cause congenial heart diseases, hypertension and also kidney problem.

In Phase III study area, water samples observed sodium concentration ranges in between 9.4 to 198 mg/l.

Potassium:

The common sources of potassium are minerals of igneous, metamorphic and sedimentary rocks. Potassium concentration in water is generally very low. BIS has not given any guideline for potassium concentration in drinking water. In the study area of Phase III, Potassium concentration observed in water samples ranged in between .08 to 35 mg/l.

The Trace Metal Concentration in thirty-one Water Samples of Phase III Study area of East Singhbhum, Saraikela-Kharsawan and Parts of West-Singhbhum Districts are given in **Annexure II**

2.4.2 Ground Water Geo-Chemistry:

Based on the major cation and anion content in the water samples and plotting them in the trilinear diagram hydrochemical facies may be identified, cation chemistry is mainly dominated by calcium followed by magnesium and sodium in the study area of Phase III. About 78% water samples are Ca-HCO₃ type in the study area. The Trilinear diagram of the study area is shown in *figure 10*.

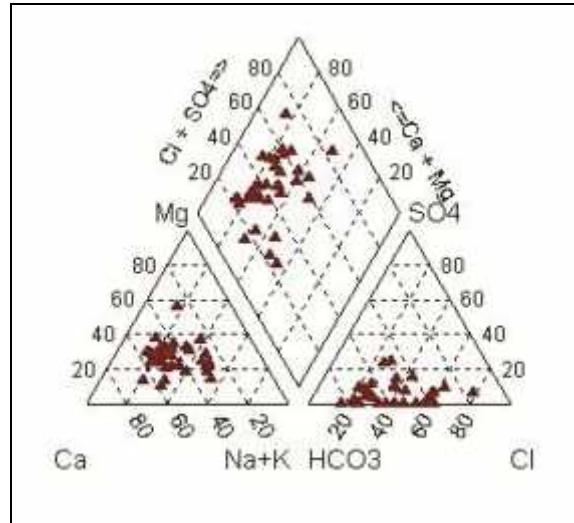


Figure 10 Piper Trilinear diagram showing Hydrochemical facies

Sodium Absorption Ratio (SAR):

In the study area of phase III all of the water samples are observed to lie in S1 & S2 classes exhibiting low sodacity. US salinity diagram of water samples of study area of phase III is shown in *figure 11*. Such water can be used on practically all kinds of soils without any risk or increase in exchangeable sodium i.e.

S1 – Low Sodium Water (SAR <10)

S2 – Medium Sodium Water (SAR 10-18)

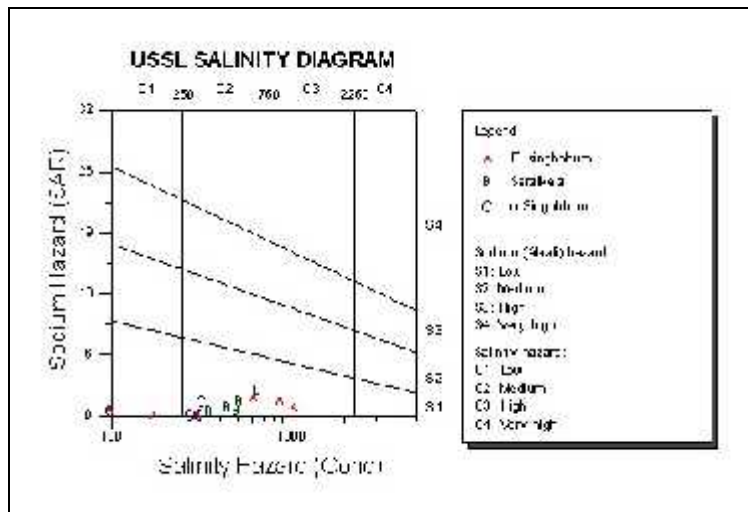


Figure 11 US salinity diagram of water samples

2.5 GEOPHYSICAL STUDIES

Under aquifer mapping studies 43 VES were conducted in East Singhbhum and Saraikela-Kharsawan, parts of West-Singhbhum districts. Out of the 43 VES, 11 VES were conducted in East Singhbhum district and 27 VES were conducted in Saraikela- Kharsawan district and 5 VES were conducted in parts of West-Singhbhum district in two major lithounits i.e. Singhbhum granite, Metasediments/Metavolcanics. The details of the VES conducted locations are given below in *Table 16*

Table 16 Details of conducted VES location in East Singhbhum & Saraikela-Kharsawan and parts of West-Singhbhum districts.

District	Site	Litho-units
East Singhbhum	Hesel (Dyke)	Singhbhum granite
East Singhbhum	Hesel (Quartz reef)	Singhbhum granite
East Singhbhum	Sonardih	Singhbhum granite
Saraikela-Kharsawan	Gitalata	Singhbhum granite
Saraikela-Kharsawan	Rola	Singhbhum granite
Saraikela-Kharsawan	Kumdih	Singhbhum granite
Saraikela-Kharsawan	Rajnagar	Singhbhum granite
East Singhbhum	Edal	Singhbhum granite
Saraikela-Kharsawan	Khairboni	Singhbhum granite
Saraikela-Kharsawan	Kudada	Meta-sediments
Saraikela-Kharsawan	Bara	Meta-sediments
Saraikela-Kharsawan	Gamharia	Meta-sediments
Saraikela-Kharsawan	Rangatad	Meta-sediments
East Singhbhum	Ulda	Meta-sediments
East Singhbhum	Mahulia	Meta-sediments
Saraikela-Kharsawan	Dugni	Meta-sediments
Saraikela-Kharsawan	Sini-Jambaha	Meta-sediments
Saraikela-Kharsawan	Janumbera	Meta-sediments
Saraikela-Kharsawan	Naryanpur	Meta-sediments
Saraikela-Kharsawan	Nilmohanpur	Meta-sediments
Saraikela-Kharsawan	Saraikela	Meta-sediments
Saraikela-Kharsawan	Ghoringa	Meta-volcanics
Saraikela-Kharsawan	Dalma	Meta-volcanics
Saraikela-Kharsawan	Nurdih	Meta-volcanics
Saraikela-Kharsawan	Kandra	Meta-sediments
East Singhbhum	Telco	Meta-sediments
East Singhbhum	Singhbhum S.Z	Meta-sediments
East Singhbhum	Boram	Meta-sediments
East Singhbhum	Potamda	Meta-sediments
East Singhbhum	AMD, Jamshedpur	Singhbhum granite
Saraikela-Kharsawan	Bhandrasai	Singhbhum granite
Saraikela-Kharsawan	Chota Dwana	Singhbhum granite
Saraikela-Kharsawan	Chowka	Singhbhum granite
Saraikela-Kharsawan	Katia	Singhbhum granite
Saraikela-Kharsawan	Kuchai	Meta-sediments
Saraikela-Kharsawan	Murkum	Singhbhum granite
Saraikela-Kharsawan	Purandih	Singhbhum granite
Saraikela-Kharsawan	Rolla	Singhbhum granite
West-Singhbhum	Bandgaon	Meta-sediments

2.5.1 GEOPHYSICAL RESULTS

Geophysical surveys in two major lithounits of the hard rock terrain were conducted at 27 sites. Out of which detailed survey were carried out at 5 sites selected for exploratory drilling. They are at Hesel in Singhbhum granites and Kudada, Ulda Dugni and Ghorlinga (metavolcanics) in metasediments.

2.5.1.a Singhbhum Granite Hesel

Near Hesel magnetic and VLF surveys were carried out across extensive quartz reef at Rola and Kumdih. The profile indicated (*figure12*) the presence of conductive overburden and the conductors, and magnetic susceptibility contrast of the granite country rock with the quartz reef. VES were conducted at a number of sites in radial directions in the dyke-infested granite area around Bahalda. Interpreted results revealed the presence of weathered zone with very low resistivity (3 Ohm.m) expected to be indicative of deterioration in quality of groundwater due to stagnation. The ellipticity of the polar diagram of VES indicates electrical anisotropy in varied directions, attributable to the presence of multi-directional structural inhomogeneities.

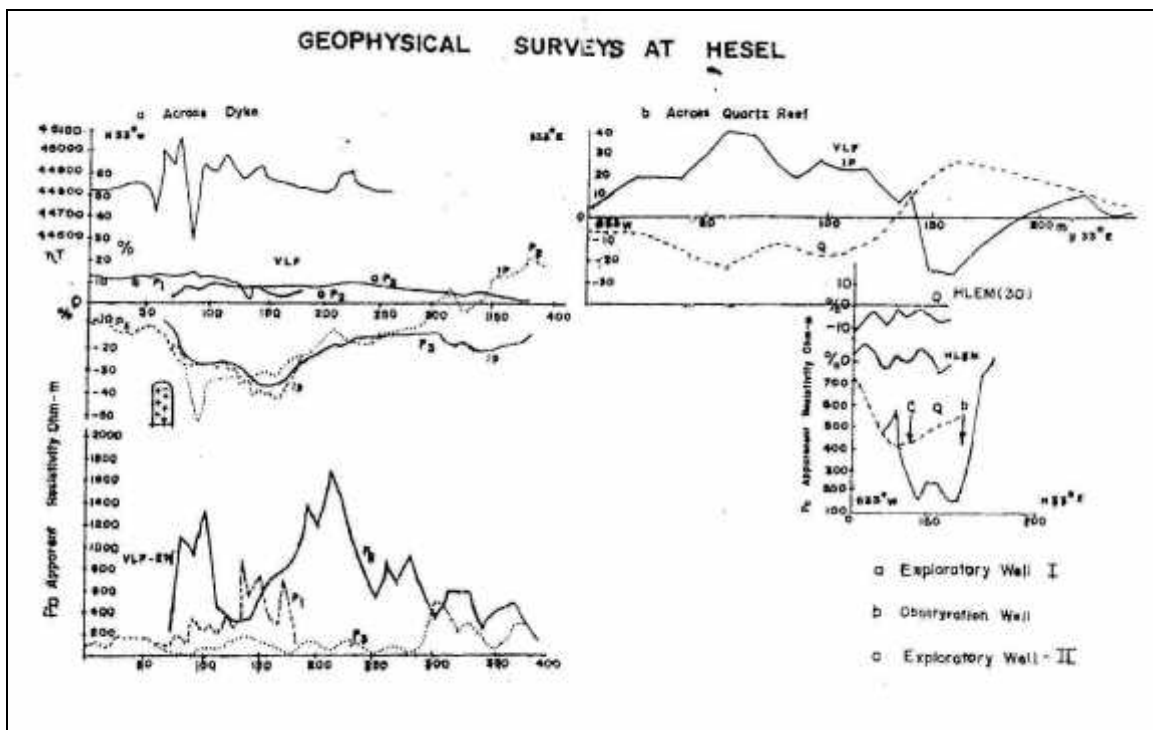


Figure 12 Geophysical survey at Hesel

2.5.1.b Metasediments

Kudada

Reconnoitory magnetic profile between kudada and nandup hill located towards north which is shown in *figure 13*, revealed the lowering in magnetic field intensity values towards the hill. The gradient resistivity profiles in the area revealed the occurrence of a flat and wide (150m) "Low" resistivity (300 to 350 Ohm.m) zone immediately towards north of kudada. They are suggestive of the deeper occurrence of fresh rocks. Two VES located on the GRP "low" indicated the presence of 35 to 40m thick weathered zone of 80 Ohm.m

resistivity. The substratum at 40m is associated with resistivity around 300 – 400 Ohm.m. Analysis of the VES curve by curve-break method revealed the presence of fracture zones in the depth ranges of 75 to 80, 85 to 90 and 95 to 100 mbgl.

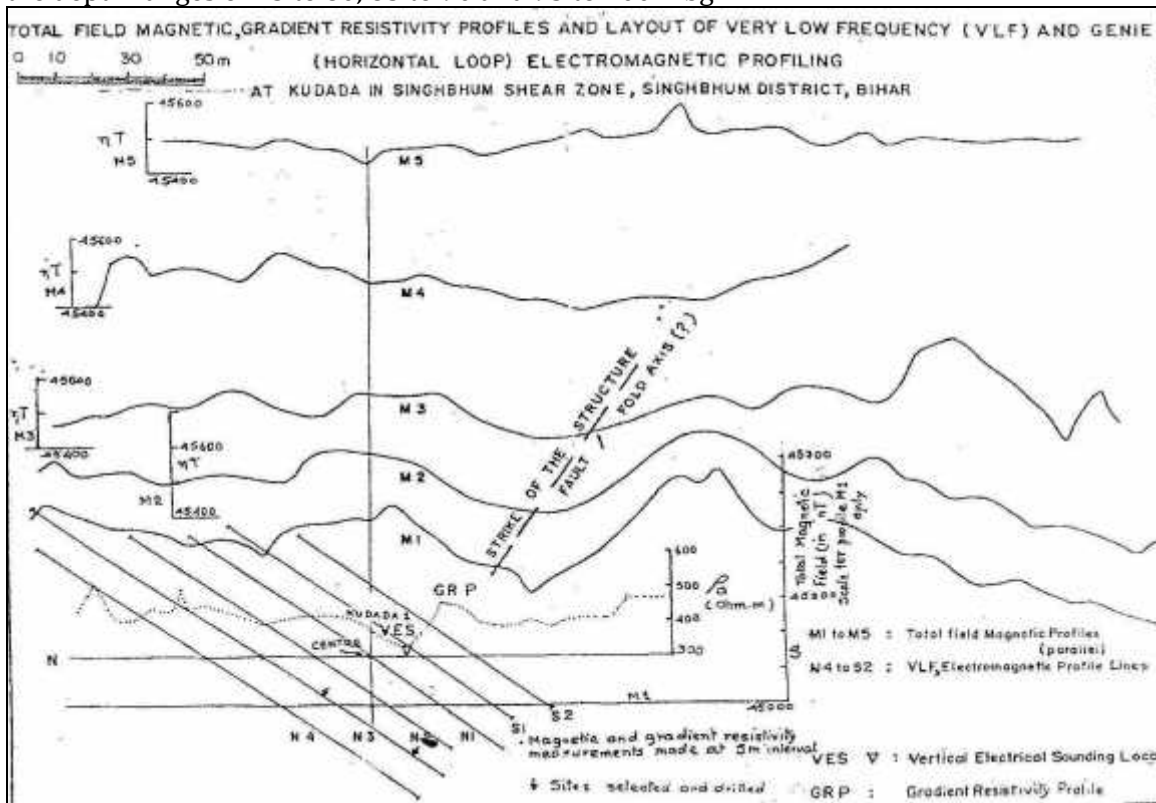


Figure 13 Electromagnetic profiling at Kudada

Ulda

Ulda-Mahulia area falls in the eastern flank of the Singhbhum shear zone. The metasediments viz. shists, quartzites and phyllites exposed in the area are highly folded with steeply dipping flanks. Two prominent NNE-SSW trending aerial photo lineaments were picked up in the area. Reconnoitry HLEM survey (P1) was carried out across the eastern lineaments near Mahulia, which is shown in figure 14. It indicated the presence of a shallow conductor. Subsequently detailed geophysical surveys along P2 and P3 were conducted in the AM low across the possible continuation of western photo lineament. The part of the magnetic profile (P2) indicated an anomaly attributable to a shallow, wide, tabular body steeply dipping towards NW of the profile. The gradient resistivity surveys (P2) indicated a step like profile towards NW with decrease in resistivity from 2000 to 500 Ohm.m. It revealed the possibility of encountering better zones towards NW. Borehole drilled at two sites located on the EM anomaly encountered productive fractures. The 64" normal resistivity log of the exploratory well revealed the presence of fracture zones at 37, 47, 56, 68 and 127m depth.

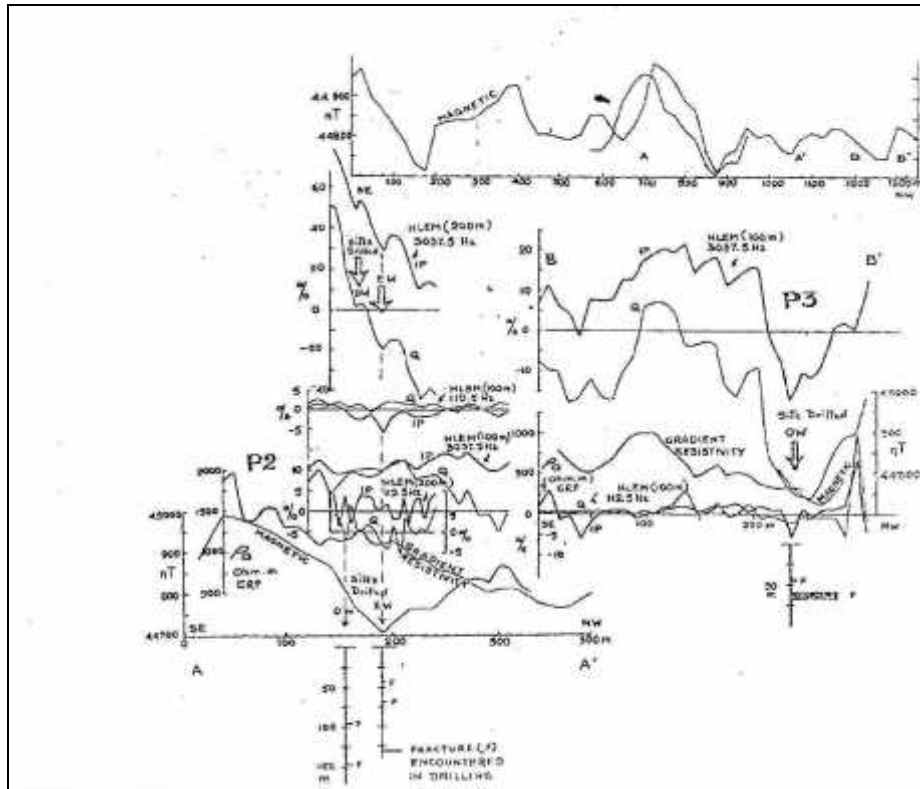


Figure 14 Reconnoitry HLEM survey at Ulda

Another profile P3 was placed about 450m northwest of P2. The lateral magnetic variation is comparatively less. The gradient resistivity low value range from 500 to 300 Ohm.m in its northwestern part. A prominent high frequency (3037.5 Hz) EM anomaly is developed in the gradient resistivity low zone. The interpreted layer parameters obtained from VES at this site are 6 to 8m thick layer of 22 Ohm.m resistivity, followed by a 35 to 40m thick layer of resistivity 60 to 70 Ohm.m. the underlying substratum is highly resistive. The borewell drilled on the anomaly encountered highly productive fractures within 70m depth.

Table 17 Result of Geophysical studies in the Granites and Meta-sediments

S.N	Location	Geology	Geophysical Techniques	Prominent Geophysical Anomaly	Weathered Zone	Remarks
1.	Hesel (Quartz Reef)	Singhbhum Granite	MAG, VLF, GRP, HLEM, VES, LOG	MAG VLF	100	GP anomaly near a dyke
2.	Kudada	Metasediments	MAG, VLF, GRP, HLEM, VES	MAG HLEM	40-80	Auto flowing well
3.	Ulda	Metasediments	AM, MAG, GRP, HLEM	AM, MAG, GRP, HLEM	20-65	
4.	Dugni	Metasediments	AM, MAG, GRP, VLF, HLEM	AM, MAG, GRP, VLF, HLEM	100	Confirmation of aerial Photo lineaments
5.	Ghorlinga	Meta volcanics	MAG, VLF, GRP, HLEM, VES, LOG	MAG, GRP, HLEM	20	Confirmation of aerial Photo lineaments and AM

AM : Aeromagnetic

MAG : Magnetic (Total field) Profiling

VLF : Very Low frequency Electromagnetic Profiling (22.3 Hz)

GRP : Gradient Resistivity Profiling

HLEM : Horizontal Loop Electromagnetic Profile (112.5, 337.5, 1012.5 and 3037.5 Hz)

LOG : Logging

District wise geophysical results are discussed below:

East Singhbhum: In East Singhbhum district Boram block office , Potmada block office and AMD Campus, Jamshedpur were referred for geophysical surveys. The result of geophysical survey is shown in *figure 15*. Details are described below:

a. Boram Block office: In the campus, 3 sites were covered with VES-27, 28 and 29. These sites are located in the premises of new Block Office complex. The VES results show three layered geo electric section. The top soil resistivity varies from 23 to 211 ohm.m with thickness of 0.8 to 2.1m. The weathered layer resistivity ranges from 10 to 90 ohm.m with a thickness of 2.3 to 12.9m. The third layer resistivity ranges from 1501 to very high indicating massive nature of the rock and is expected at a depth range of 4.4 to 13.7m. At VES-28 within the massive rock fractures are expected at depth.

b. Potmada Block office: In the campus, 2 sites were covered with VES-25 and 26. These sites are located in the premises of Block Office campus. The VES results show three and four layered geo electric section. The top soil resistivity varies from 207 to 263 ohm.m with a thickness of 0.8 to 1.8m. At VES-25 the second layer resistivity is 90 ohm.m with thickness 3.5m which is absent at VES-26. The weathered layer resistivity is around 24 ohm.m with a thickness of 3.0 to 4.0m. The third layer resistivity ranges from 3373 to very high indicating massive nature of the rock and is expected at a depth range of 4.8 to 8.3m. At VES-26 within the massive rock fractures are expected at depth.

c. AMD Campus, Jamshedpur Block: In this campus, 5 sites were covered with VES-53 to 57 and 100m GRP. The GRP conducted with AB=200M and MN=10m with station interval of 5m. The profile indicates not much variation up to 40m, then decreases 40-50m and increases up to 90m. These sites are located in the premises of AMD campus. The VES results show three and four layered geo electric section. The top soil resistivity ranges from 23 to 496 ohm.m with a thickness of 0.8 to 2.0m. The weathered /semi weathered layer resistivity varies from 75 to 196 ohm.m with a thickness of 2.9 to 44.7m. At VES-56 a layer with a resistivity of 40 ohm.m with thickness 4.1m which is absent at other VES. The third layer resistivity ranges from 369 to very high and is expected at a depth of 7.9 to 45.8m. At VES-53, 55 and 57 within the massive rock fractures are expected at depth.

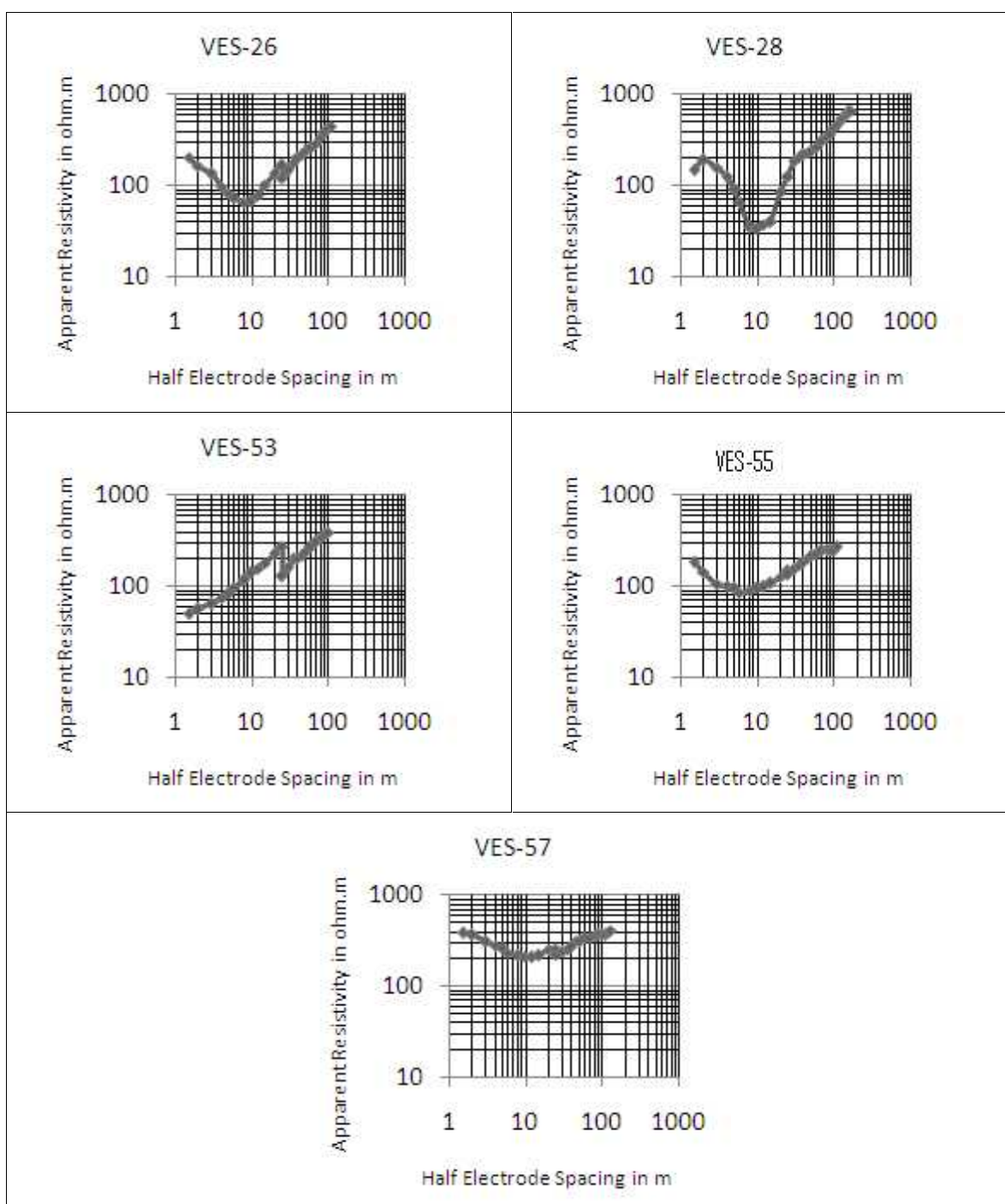


Figure 15 VES Curves of Recommended drilling sites in Potmada, Boram and AMD Campus, Jamshedpur of East Singhbhum District

Saraikela- Kharswan: In the study area of (Phase III) Sarikela-Kharsawan district Bhandrasai, Chota Dawana, Chowka, Katia, Kuchai, Murkum, Purandih and Rolla Villages were referred to geophysical surveys. The result of geophysical survey is shown in *figure 16*

a. Bhandrasai, Saraikela Block:

In this village, three sites were covered with VES-38, 39 and 40. These sites are located in the premises of school play ground. The VES results show three to four layered geo electric section. The top soil resistivity varies from 124 to 148ohm.m with a thickness of 0.8 to 4.2m. At VES-38 this is followed by 12ohm.m resistive layer with thickness 1.0m. The

second layer resistivity varies from 43 to 79ohm.m with a thickness of 6.3 to 15.5m. The last layer resistivity is very high and is expected at a depth range of 8.2 to 19.7m. At VES-39 with in the massive rock fractures are expected at depth.

b. Chota Dawana, Saraikela Block:

In this village three sites were covered with VES-35, 36 and 37. These sites are located in the premises of Panchayat office campus. The VES results show three to four layered geo electric section. The top soil resistivity varies from 18 to 59ohm.m with a thickness of 0.9 to 1.6m. The second layer resistivity varies from 11 to 50ohm.m with a thickness of 1.1 to 9.5m. At VES-35 this is followed by 232ohm.m resistive layer with thickness 20.7m. The last layer resistivity is very high and is expected at a depth range of 11.1 to 23.6m.

c. Chowka, Chandil Block:

In this village three sites were covered with VES-16, 17 and 18. These sites are located in the premises of school play ground. The VES results show three to four layered geo electric section. The top soil resistivity varies from 21 to 46ohm.m with a thickness of 1.6 to 3.4m. At VES-16 this is followed by 54ohm.m resistive layer with thickness 1.4m. The second layer resistivity varies from 5 to 12ohm.m with a thickness of 4.1 to 8.7m. The last layer resistivity is very high and is expected at a depth range of 7.1 to 10.9m.

d. Katia, Chandil Block campus:

In this campus, three sites were covered with VES-22, 23 and 24. These sites are located in the premises of Block office campus. The VES results show three layered geo electric section. The top soil resistivity varies from 20 to 23ohm.m with a thickness of 0.8 to 1.6m. The second layer resistivity varies from 2 to 7ohm.m with a thickness of 0.9 to 4.8m. The last layer resistivity is very high and is expected at a depth range of 2.4 to 5.6m.

e. Kuchai, Kuchai Block campus:

In this campus, three sites were covered with VES-50, 51 and 52. These sites are located in the premises of stadium campus. The VES results show three to four layered geo electric section. The top soil resistivity varies from 84 to 135ohm.m with a thickness around 0.9m. The second layer resistivity varies from 13 to 15ohm.m with a thickness of 3.5 to 11.7m. The third layer resistivity varies from 6 to 89ohm.m with a thickness of 5.5 to 21.6m. The last layer resistivity varies from 645ohm.m to very high and is expected at a depth range of 10.0 to 25.9m. At VES-51 with in the massive rock fractures are expected at depth.

f. Murkum, Saraikela Block:

In this village, three sites were covered with VES-47, 48 and 49. These sites are located in the premises of Panchayat office campus. The VES results show three to four layered geo electric section. The top soil resistivity varies from 82 to 429ohm.m with a thickness of 0.8 to 1.9m. At VES-48 this is followed by 93ohm.m resistive layer with thickness 2.4m. The second layer resistivity varies from 16 to 37ohm.m with a thickness of 3.2 to 6.6m. The last layer resistivity is very high and is expected at a depth range of 6.4 to 8.5m.

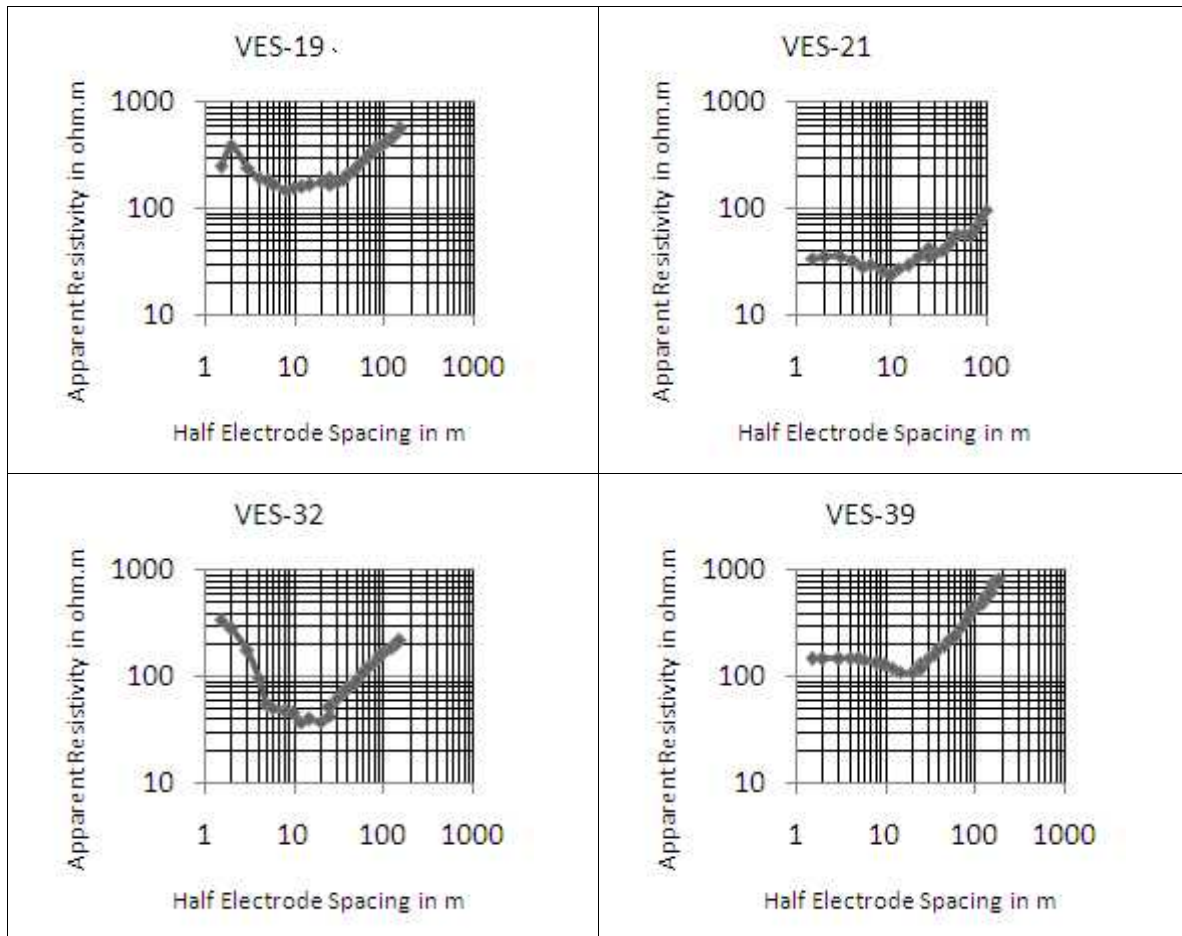
g. Purandih, Ichagarh Block:

In this village three sites were covered with VES-19, 20 and 21. These sites are located in the premises of School campus. The VES results show three to four layered geo electric section. The top soil resistivity varies from 31 to 299ohm.m with a thickness of 1.4 to 3.3m. At VES-20 and 21 this is followed by a layer with resistivity 8-42ohm.m and

thickness 1.9-2.6m. The second layer resistivity varies from 62 to 128ohm.m with a thickness of 20.3 to 53.7m. The last layer resistivity is very high and is expected at a depth range of 21.7 to 57.2m.

h. Rolla, Rajnagar Block:

In this village three sites were covered with VES-32, 33 and 34. These sites are located in the premises of Kartik Oraon High School campus. The VES results show three to four layered geo electric section. The top soil resistivity varies from 78 to 135ohm.m with a thickness of 0.9 to 1.7m. At VES-34 this is followed by 9ohm.m resistive layer with thickness 0.7m. The second layer resistivity varies from 31 to 51ohm.m with a thickness of 9.2 to 14.5m. The last layer resistivity is very high and is expected at a depth range of 10.1 to 16.7m. At VES-32 with in the massive rock fractures are expected at depth.



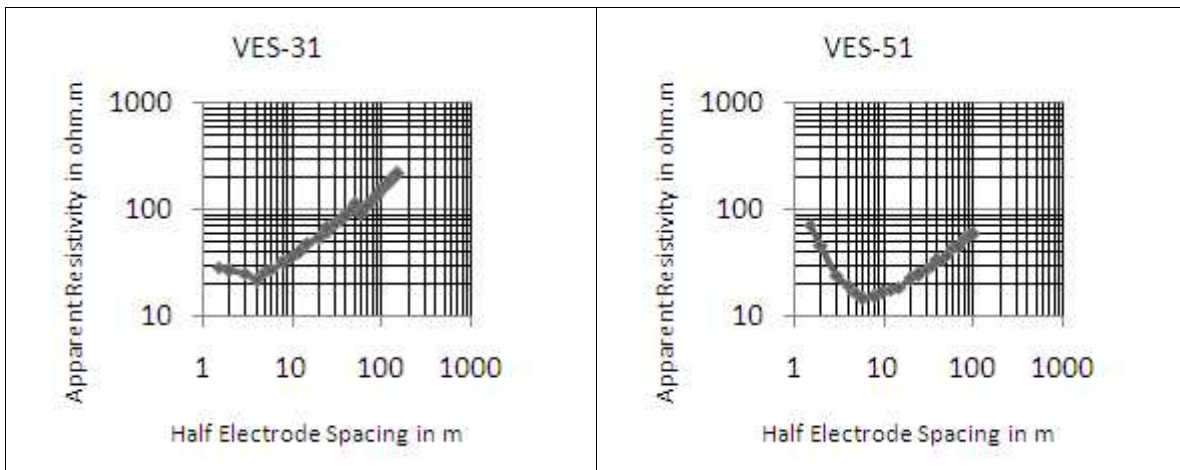


Figure 16 VES Curves of Recommended drilling sites in Purnadih, Rolla, Pandrasalai, Haldi Pokar and Kuchai of Saraikela-Kharswan District.

West Singhbhum District: In this district geophysical survey was carried out in Bandgaon area.

a.Bandgaon, Bandgaon Block:

In this village three sites were covered with VES-13, 14 and 15. The VES-13 is located in the premises of Block office campus. The VES-14 and 15 sites are located in the premises of Stadium campus. The VES results show three to five layered geo electric section. At VES-13 the top soil resistivity is 117ohm.m with a thickness of 0.9m. The second layer resistivity is 41ohm.m with a thickness of 9.2m. The last layer resistivity is 632ohm.m and is expected at a depth of 10.1m. The VES results at stadium campus show four to five layered geo electric section. The top soil resistivity varies from 225 to 489ohm.m with a thickness of 0.9 to 1.8m. The second layer resistivity varies from 412 to 725ohm.m with a thickness of 1.2 to 6.9m. The third layer resistivity varies from 148 to 263ohm.m with a thickness of 2.6 to 8.7m. At VES-14 this is followed by 394ohm.m resistive layer with thickness 20.1m. The last layer resistivity is very high and is expected at a depth range of 17.4 to 24.8m. The result of geophysical survey is shown in *figure 17*.

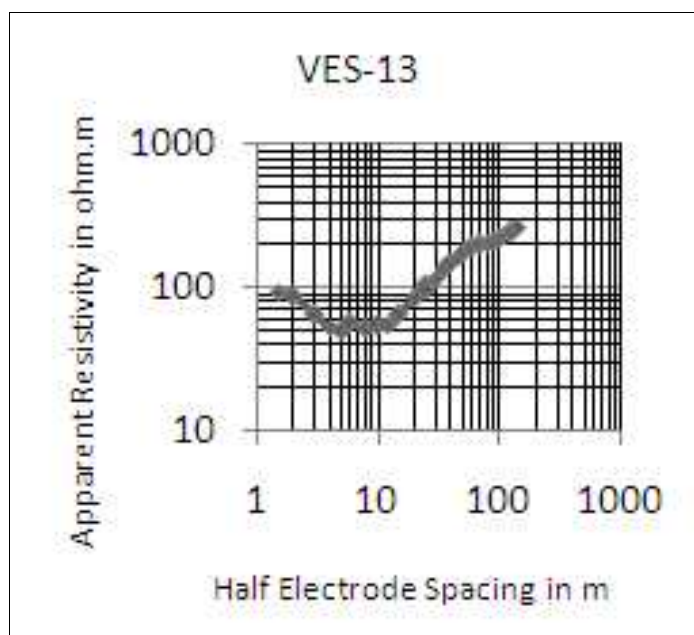


Figure 17 VES Curves of Recommended drilling sites in Bandgaon of West Singhbhum District.

Based on VES results, the resistivity characteristics of different Granitic rocks are furnished in Table 18

Table 18 Resistivity Characteristics inferred from VES in Granite rocks.

Resistivity in Ohm.m	Thickness in m	Expected Rock type
18-996	0.8-7.7	Surface Soil
2-19	0.6-11.0	Highly weathered rock
20-100	1.4-39.0	Weathered/Semi Weathered rock
105-434	2.0-57.0	Fractured rock
570-1500	2.0-66	Minor Fractured rock
>2000	-	Massive rock

2.6 GROUND WATER EXPLORATION:

The exploratory drilling programme was taken up with the aimed to identify and delineate aquifers, decipher geometry of the fracture system and their hydraulic continuity and also determine the aquifer parameters and yield characteristics. With the above objective in view, CGWB has drilled 41 no. of exploratory well in the NAQUIM Phase III area of different lithounits. The location maps of exploratory wells are shown in the *figure 18*. Exploratory drilling in the area was carried out to the maximum depth of 209 mtr. The area is underlain by the 3 types of lithounits namely granite rocks (chakardharpur granite/Singbhum granite), metasediments and metavolcanics with isolated occurrences of sandstone, shale and carbonate rocks of Precambrian age. The granitic rocks show wide variations from Singhbhum granite to Chakardharpur granite. The metasedimentaries also shows variations in texture, structure and age. Among the intrusive, basic and meta basic Intrusives are most common and these are predominantly found within the singhbhum granitic rocks in a criss-cross fashion. The occurrence and movement of ground water are mainly controlled by the litho-structural characteristics of underlying formations. Three major hydrogeological units are encountered in the hard rock terrain, which contribute

ground water reservoir i.e weathered mantle, saprolite zone and fracture, fissure and joints. The hydrogeological conditions of this different ground water bearing zones are detailed below:

Weathered zone:

Ground water occurs in the weathered zone under phreatic condition. It is the most extensive near surface aquifer. the thickness of the weathered zone as well as its water bearing and water yielding properties vary depending on the geomorphic and geological set up both locally as well as regionally. It is observed that in granite gneiss weathered zones depth ranges between 1-40 mbgl considering first aquifer. depth to water level ranges 3.65-12 mbgl, yield of the well ranges 0.45 to 5 lps.

Saprolite zone:

The transition zone between the weathered mantle and fresh rock has been designated as saprolite, which is persistent throughout the area. The saprolite zone are well developed in the Chakardharpur granite with the average thickness around 5 mtr while in Singhbhum granite the average thickness of this zone is around 3.5 mtr only. The average thickness in the metasedimentaries is around 4 mtr. The yield from this zone varies from 0.45 to 2.66 lps.

Fracture/Fissure/Joints zones:

These are encountered within the depth 40-200 mbgl. The depth to water level ranges between 3.3-20 mbgl and yield ranges 0.45 to 11.5 lps. Majority of fractures are encountered within 40-120 mbgl in Granitic Rocks. In Granite few fractures are encountered within 160-185 mbgl.

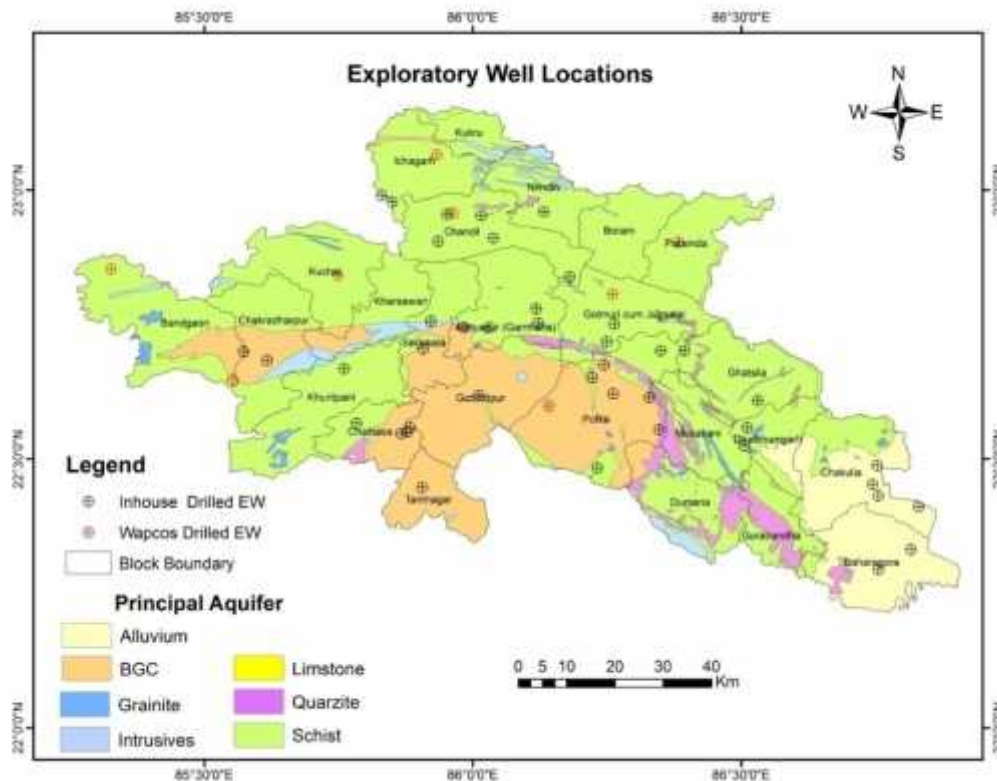


Figure 18 Exploratory well Location map

3.0 DATA INTERPRETATION, INTEGRATION AND AQUIFER MAPPING

Based on exploratory wells data, lithological section(2-D and 3-D) have been prepared. The lithological section map is shown in *figure 19*. Three types of major formations are in the study area i.e. Granitic rock, Metasediments /Metavolcanics & Teriary. In granite-gneiss terrain lithological section A-A' which is along Sildaauri Middle School, EW to Nimdih exploratory well having maximum one fractures were encountered which is shown in *figure 20* while in section B-B' prepared in between Purandih School campus, Ew to Uliyan Kadma Ew here minimum one set and maximum four set of fracture were encountered, which is shown in *figure 21*. Maximum numbers of fractures are encountered in Schist. In section C-C' minimum one and maximum six set of fracture encountered. Maximum no of fracture i.e. six set of fracture encountered in Hesel EW which is shown in *figure 22* In section D-D' having one to three number of fracture zones were encountered which is shown in *figure 23*

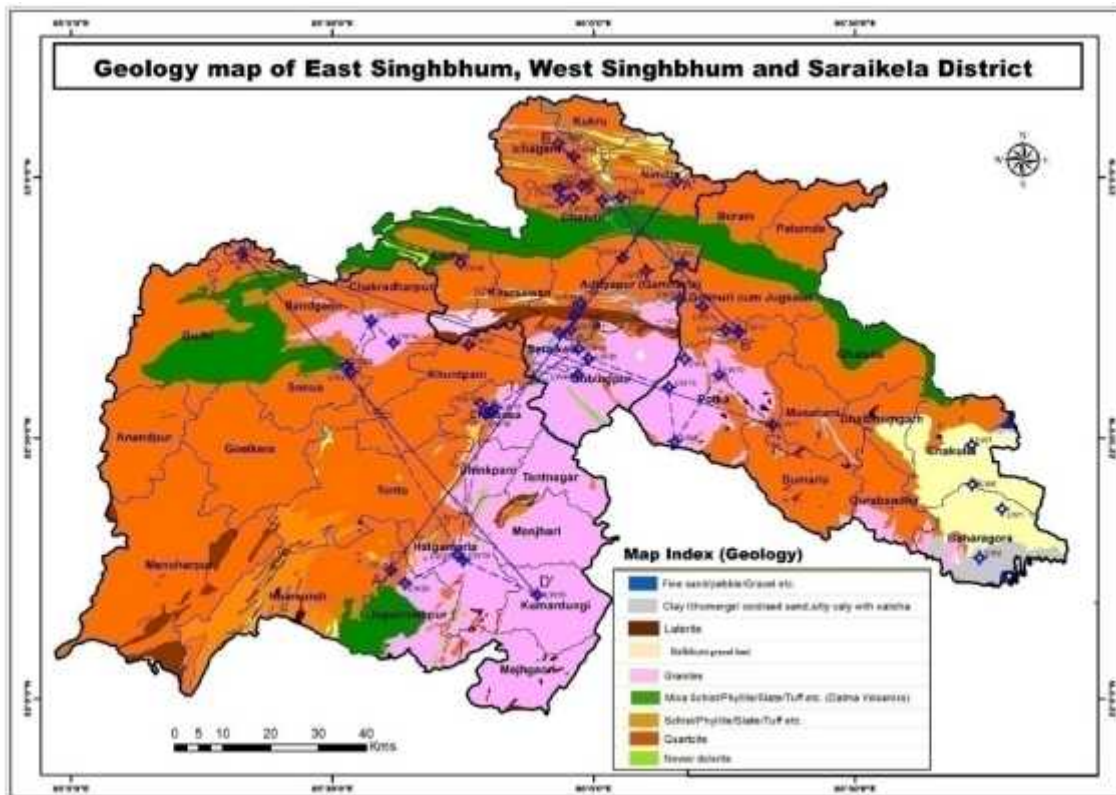


Figure19- Lithological section

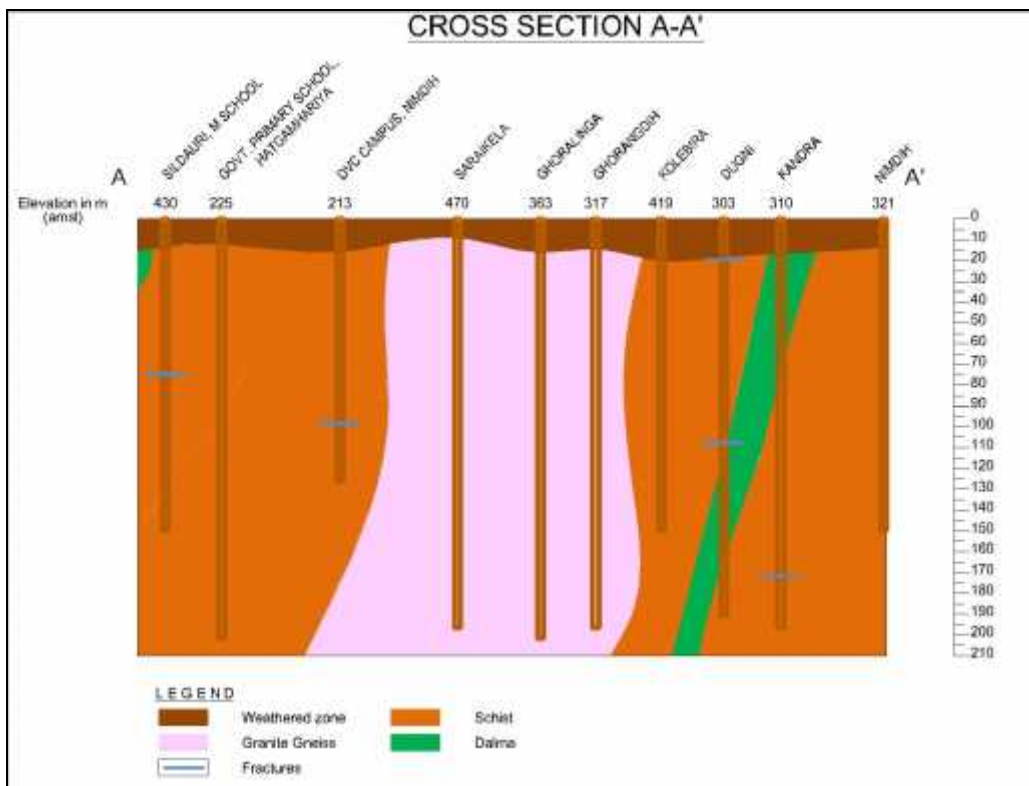


Figure 20 Lithological cross-section A-A'

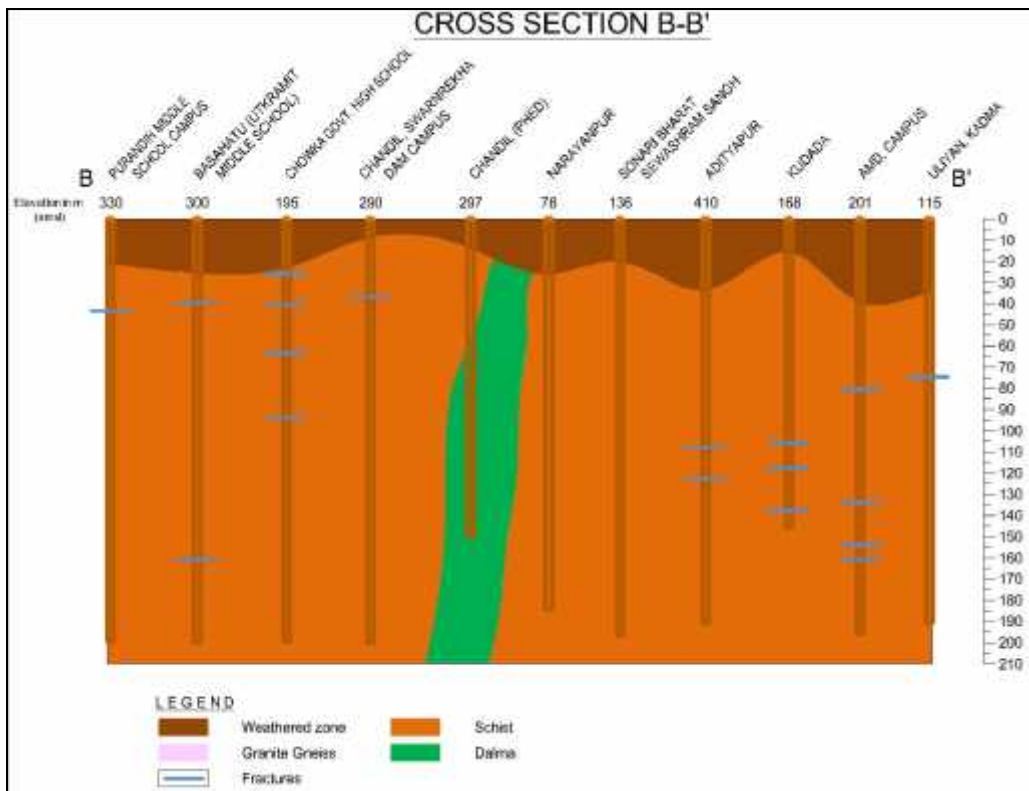


Figure 21 Lithological cross-section B-B'

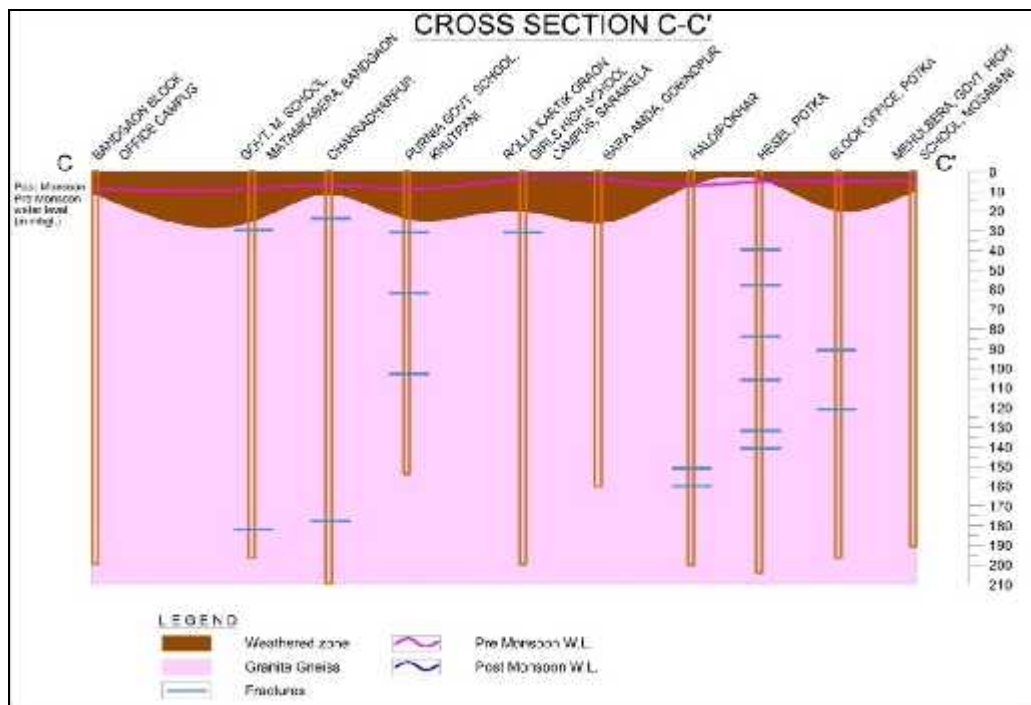


Figure 22 Lithological cross-section C-C'

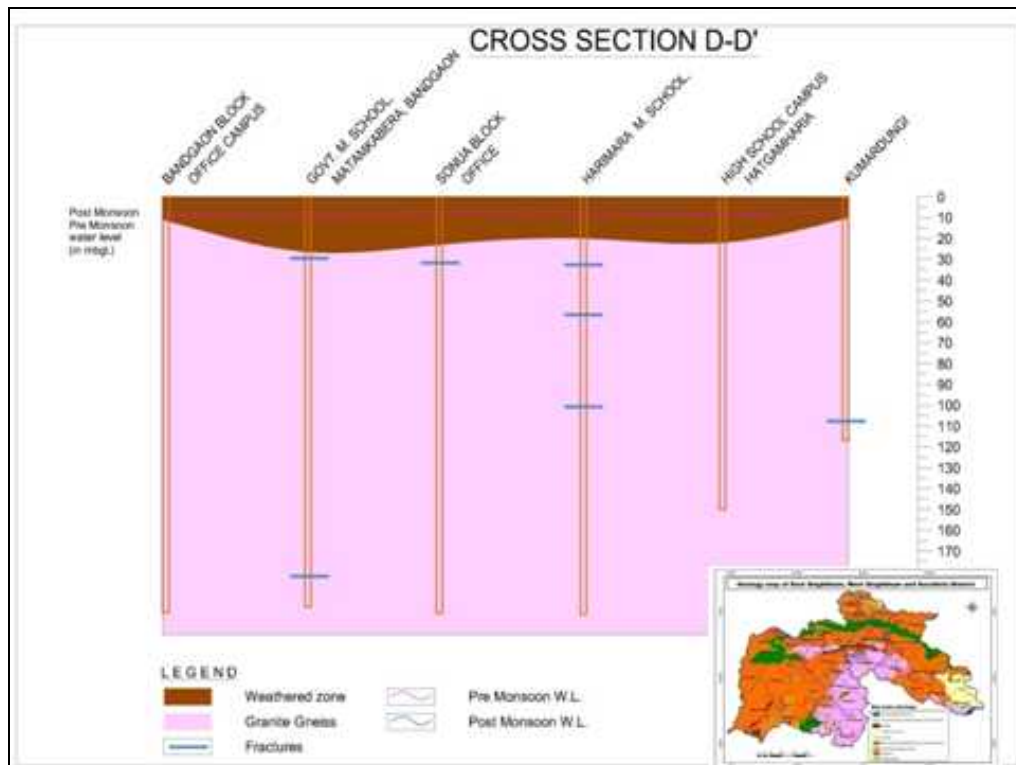


Figure 23 Lithological cross-section D-D'

Analysis of Fracture System:

The analysis of fracture system refers that there are two aquifer systems lies in phase III study area, which is classified by its depth ranges i.e. Aquifer I & II. Depth of aquifer I is generally 0 to 30 mtr which is weathered and aquifer II depth varies from 30 to 200 mbgl. The details of fracture system in given in Table 19

Sl. no.	District	Well Drilled		Dry	productive fracture	Depth range(m)	Yield (lps)
1	East-Singhbhum	Tertiary (Alluvium)	4	nil	Mixed granular zone	0-120	15-21
		Metasedimentary	13	7	5	0-100	3.6
					7	100-200	
		Schist (Singhbhum Group)	8	8			
2	West-Singhbhum	Chakradharpur Grainite	1	nil	1	0-100	< 2
					1	100-200	
		Schist (Singhbhum Group)	8	7	2	0-100	< 2
					1	100-200	
3	Saraikela	Metasedimentary	4	nil	5	0-100	< 1
					2	100-200	
		Schist (Singhbhum group)	2	2			
		Basic rocks (Dalma)	1	1			
	Total		41	25			

[illegible]

33

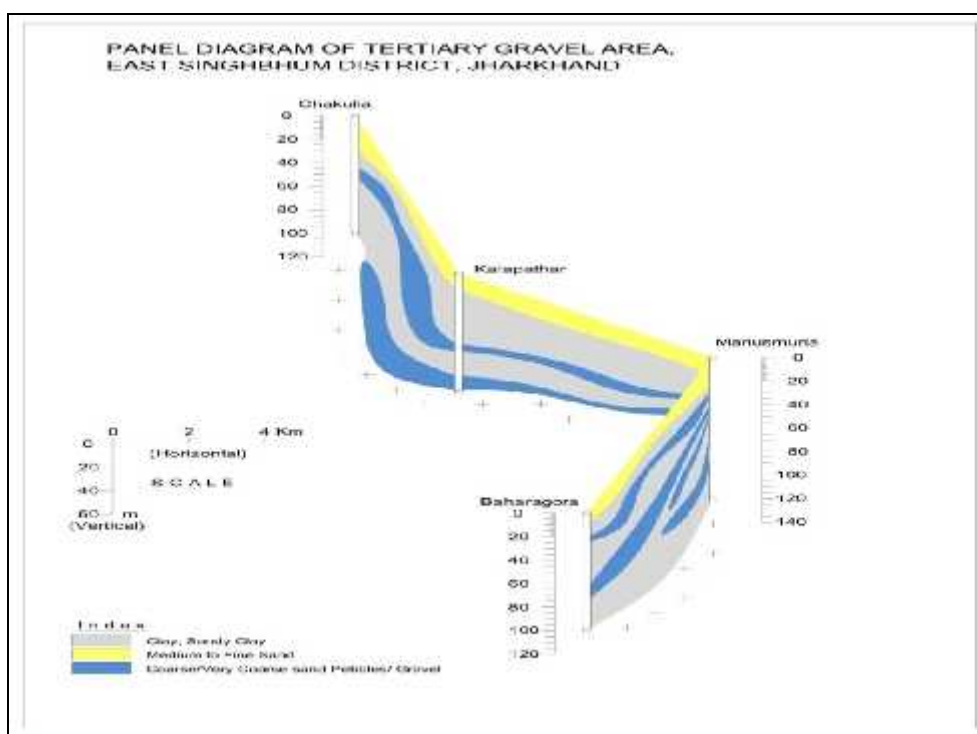


Figure 25 Panel Diagram of Tertiary, Gravel area of East Singhbhum District

4.0 GROUND WATER RESOURCES

Ground Water Resource of the area has been estimated based on block wise for base year as on March-2013. There are twenty five number of assessment units (blocks) in the study area where one block come under over-exploited namely Golmuri-cum-Jugsalai. Out of twenty five blocks twenty four blocks falls in safe categories. The net dynamic ground water availability in the area is 48667.55 ham and ground water draft for all uses is 9188.63 ham, making stage of ground water development 19% as a whole for the phase III area. After making allocation for future domestic and industrial water supply for next 25 years, balance available ground water for future irrigation development would be 44315.8 ham. The details of ground water resource of study area is given in *Table 20*

Table 20 Details of Ground Water Resource (As on March -2013)

S.no	Items	East Singhbhum	Saraikela-Kharsawan	West Singhbhum
1	Area in ham	356200	265900	168400
2	Net Ground Water Availability in ham	29583.10	12351.41	6733.04
3	Existing Gross Ground Water Draft for Irrigation in ham	1692.75	753.00	207.00
4	Existing Gross Ground Water Draft for Domestic use in ham	3308.36	1538.40	674.3
5	Existing Gross Ground Water Draft for Industrial use in ham	817.65	197.17	00.00
6	Existing Gross Ground Water Draft for all Uses in ham	5818.76	2488.57	881.3
7	Net Ground Water Availability for future Irrigation Development in ham	27026.35	11051.41	6238.04
8	Stage of Ground Water Development (%)	19%	20%	13%

5.0 GROUND WATER RELATED ISSUES

5.1 Identification of Issues:

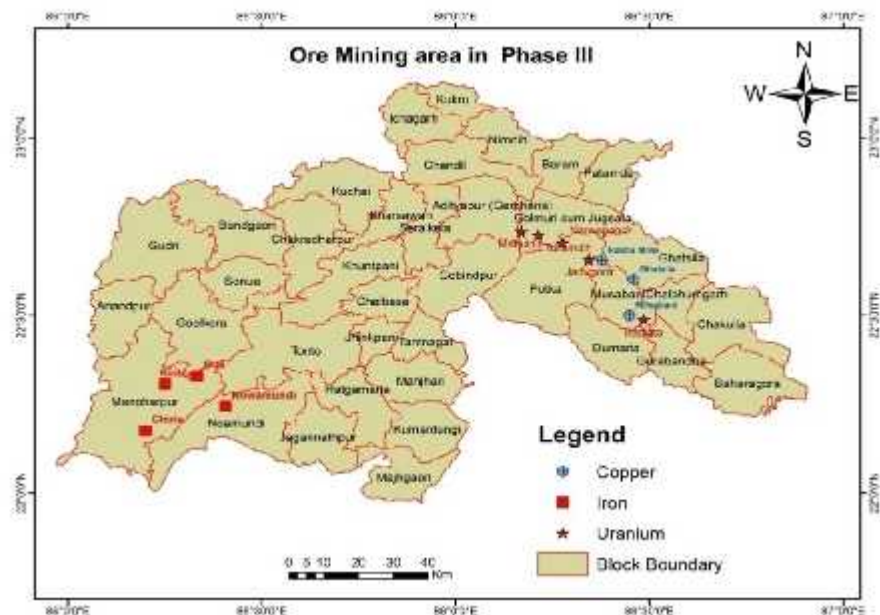
The major ground water related issues of the study area are grouped into following broad categories:

- Area of intensive mining activities
- Quantifying aspects
- Quality aspects
- Administrative issues

A variety of nature's factors affect the quantity and quality aspects of ground water over space and time. The major ground water related issues are discussed as follows;

5.1.a Area of intensive mining activities;

The study area of phase III is highly mineralized zone therefore lots of mining industries like Iron, Copper and Uranium mines are situated here. The location map of such industries is shown in *figure 26*. It is mainly associated with the mining activity with the intensive development of ground water. These are identified for the problem of depletion in the general water table and decline in the bore well discharge. Underground and opencast excavations behave as large sinks and create hydraulic gradient towards the mines. Mine water is pumped out for trouble free mining operations. Continuous withdrawal of water from mines for their mining activities is causing adverse impact on ground water regime of the area which ultimately results in declining ground water levels, drying up of wells, dwindling of their discharge and sometimes land subsidence occurs. The main ground water problem in the mining area is related with the dewatering of deeper aquifers (underground mines) i.e Uranium and Copper mines of the area. Continued pumping from these mines causes depletion of water levels.



Figure

Location map of Ore Mining area

26

5.1.b Quantity Aspects:

Ground water potential at any area mainly depends on the topography, rainfall, and geology. Because of plateau topography and Chotanagpur granitic gneiss complex as the litho-units occurring in the study area, the ground water potential is not uniform and it changes from one area to another. The salient features of ground water resources as on March-2013 is given in *Table 21*.

Table 21 Details of Ground Water Resource (As on March -2013) of the study area

S.No.	Items	Ground water in ham
1	Total annual replenishable ground water resource	54013
2	Natural losses	5401
3	Net Dynamic Ground Water Availability	48667
4	Existing Gross Ground Water Draft for irrigation	2652
5	Existing Gross Ground Water Draft for Domestic use	5521
6	Existing Gross Ground Water Draft for Industrial Water Supply	1014
7	Existing Gross Ground Water Draft for all Uses	9188
8	Provision for domestic and industrial requirement supply to next 25 years	1699
9	Net Ground Water Availability for future Irrigation Development	44315
10	Number of blocks	25
11	Number of semi-critical blocks	0
12	Number of critical blocks	0
13	Number of over-exploited blocks	1
14	Number of safe blocks	24
	Stage of development	19%

The availability of water resources is not uniformly distributed over space and time. This resource depletes often in summer, Therefore reduction of bore well's yield in lean period. The area is covered by Chotanagpur granite gneiss complexes where ground water potential of deeper aquifer is very less and limited thickness of fracture/joints are encountered in exploratory well drilling programme.

In Golmuri-cum-Jugsalai block of East Singhbhum district, high dependency prevails on ground water for Domestic purposes. Existing gross ground water draft for domestic water use is 1451.37 ham and existing gross ground water draft for irrigation is 15.75 ham. The total ground water draft for all uses is 2218.79 ham, making stage of ground water development 147.42%. Deeper aquifers of Golmuri-cum-Jugsalai block is showing declining trend of water levels. In Saraikela-Kharsawan and parts of West-Singhbhum districts there is ample scope of ground water development.

5.1.c Quality Aspects:

The ground water quality of the study area is potable and is suitable for irrigation. However at few localities geogenic contamination in ground has been observed. The major problems and issues related to the quality is related to iron contamination.

5.1.d Unscientific Development of Ground Water Resources:

The study area experiences both very high as well low level of ground water development. In the north eastern part of the area the ground water resources are over utilized whereas western and some part of central part ground water resource is underutilized. Ground water related issues and problems are not getting scientific attention of the Government, planners and stakeholders. The plan and policy of the water resources department are mostly related to the surface water only. There is urgent need to pay attention towards ground water also. The laws of compulsory Rain Water harvesting should be enacted in Nagar Nigam /Nagar Palika's of the study area. A separate pricing policy for the bulk consumers of ground water in the area should be formulated.

5.2 Future demand scenario and stress aspects of the aquifer:

Ground Water Modeling is required to estimate the different scenario and stress aspects of the aquifer response. Few of the prospective scenarios is given in *Table22*.

Table 22 Future demand scenario and stress aspects of the aquifer

S,N o,	Scenario (stress applied)	Aquifer-I Response	Aquifer-II Response	Remarks
1.	Existing Draft due to pumping			
2.	Ground water draft increased to 10% in next year			
3.	Draft increased 50% from the present level @ 5% annually from the present condition			
4.	Drought condition			

5.3 Participatory ground water management issues:

The most of the study area falls under rural and tribal population. Illiteracy, money crisis and ignorance are main reasons for under development of ground water resources. The erratic electric supply, the problem of low voltage and unavailability of electrification is hampering agricultural activities in the area..

6.0 MANAGEMENT STRATEGIES

In management part of phase III area, we have to prepare plan for over-exploited block based on the ground water resource availability and draft condition of the area.

6.1 Management objective and option:

Prime Management objective is keeping in view of stage of development 70% and Management options are ;

- o Proper utilization of mine water
- o Implementation of Rain Water Harvesting & Artificial Recharge Techniques.
- o Utilization of surface water instead of ground water
- o Ground Water Development in most irrigation deprived area i.e low ground water developed blocks.

6.2 Management plan for Over-Exploited block:

In phase III area, Golmuri-cum-jugsalai block comes under over exploited block. In Golmuri-cum-jugsalai block having intensive domestic draft rather than industrial and irrigation draft. The ground water requirement pattern of this over-exploited block should be managed to keep stage of development 70%. The brief descriptions of management plan for OE block are discussed below;

Golmuri-Cum-Jugsalai (Jamshedpur):

Ground Water requirement pattern to maintain stage of development at 70% of over-exploited block the volume of water required is given in *Table 23*

Table 23 -Details of Ground Water requirement Pattern in over-exploited block

S.no.	Items	Golmuri-Cum-Jugsalai
		Volume of water in ham
1.	Net dynamic ground water availability	1505.10
2	Existing gross ground water draft for irrigation	15.75
3	Existing gross ground water draft for domestic use	1451.37
4	Existing gross ground water draft for industrial use	751.67
5	Existing gross ground water draft for all uses	2218.79
6	Proposed gross ground water draft for all uses	1053.57
7	Reduction in ground water draft from domestic & Industrial for stage of development 70%	1165.22

Roof-Top Rain Water Harvesting:

Implementation of Roof top rain water harvesting we can also enhance the ground water availability. A generalized plan for roof top rain water harvesting is given below:

- One rooftop having area 500 square feet = 45 square meters
- Rainwater that can be harvested from one roof with 75% efficiency = 34 cubic meters

Considering urban households around 1 Lac, total rainwater that can be harvested = **3.4 million cubic meters (mcm)** can be recharge by roof top rain water harvesting system

Proposed Management Strategy for Golmuri-Cum-Jugsalai (Jamshedpur):

In the study area, Golmuri-Cum-Jugsalai which comes in overexploited block where existing gross ground water draft for domestic water use is 1451.37 ham and existing gross ground water draft for irrigation is 15.75 ham. The total ground water draft for all uses is 2218.79 ham, making stage of ground water development 147.42%. When assessing the draft for Golmuri-Cum-Jugsalai block, it is observed that the Domestic ground water draft contribute towards higher as compared to the irrigation and industrial draft.

To reduce domestic draft and keep stage of development at 70% ground water draft is to be reduced to 1165.22 ham or adopt artificial recharge practices of ground water.

6.2.a. Aquifer wise space available for recharge and proposed interventions

The computational procedure for computing the aquifer wise in-storage ground water resource comprising the following steps

- The area of the ground water assessment unit for which the static ground water resource is to be computed is obtained.
- The depth below ground level up to which the zone of water table fluctuation occurs is obtained. This can be taken as the maximum depth below ground level recorded during the pre-monsoon interval. An acceptable depth below ground level up to which ground water can be mined is obtained. This will be usually less than the maximum depth below ground level up to which the saturated aquifer formation extends.

The assessment of total availability of ground water resource has been carried out using following methodology.

In- Storage Ground Water Resource Assessment:

Thickness of Aquifer (granular/productive zone) * Sp.yield * Area below zone of fluctuation water

Parameter used: Specific Yield

Depth of Assessment of in-storage Ground Water Resources

For Alluvial Area: Up to bed rock

For Hard rock area: Depth up to which the aquifer is commonly developed

6.2.b Aquifer wise ground water resource availability of Golmuri-Cum-Jugsalai:

The over-exploited area comes under hard rock terrain therefore aquifer wise in-storage ground water resource availability were calculated considering specific yield 1.5% for aquifer I and specific yield 1% for Aquifer II. Aquifer-wise In-storage ground water resource is calculated below:

For Aquifer I in hard rock area of Golmuri-Cum-Jugsalai block, district East-Singbhum

AQUIFER I	
Area (A) (sqkm)	203
Pre-monsoon (average) depth to water level (mbgl) (B)	12.08
Average depth to bed rock (mbgl) (C)	30
Specific yield (Sy)	1.5%
Saturated zone thickness (B-C) of aquifer (ST)	17.92
Resource (A * Sy * ST) (20300ha * 0.015 * 17.92)	5456 mcm

For Aquifer II in hard rock area of Golmuri-Cum-Jugsalai block, district East Singbhum

AQUIFER-II	
Area (A) (sq km)	203
Thickness of fracture from weathered zone to the depth of 200m	3 m
Specific yield (SY)	1%
Resource (A * SY * ST) (20300Ha * 0.01 * 3)	609 mcm

Aquifer wise statement for hard rock of Golmuri-Cum-Jugsalai block, East Singbhum

Aquifer	In-Storage Ground water resource
AQUIFER-I	5456 mcm
AQUIFER-II	609 mcm

Total Availability = Dynamic Ground Water Resource + In-Storage Ground Water Resource

$$\text{Total Availability (Mcm)} = 15.05\text{mcm} + 6065\text{ mcm} = 6080\text{mcm}$$

6.2.c Estimation of Available Sub Surface Storage Potential/ Source Water requirement/availability for recharge, Types and number of Recharge structure

The thickness of unsaturated zones (below 3 mbgl) of above categories is estimated by considering the different range of water level. The total volume of unsaturated zone is calculated by considering the different water level zones and unsaturated thickness in different ranges of water level. This volume was multiplied by specific yield i.e. 3 % for hard rock on area specific basis to arrive at the net amount of water required which is to be recharged by artificial recharge to saturate the aquifer up to 3 mbgl. The details of sub surface storage potential and number of AR structures required in the study area is given in Table 24

Table-24 Details of sub surface storage space available & number of AR structures required for – Hard rock

Golmuri-Cum-Jugsalai	Sub surface storage available (mcm)	Surface water required (mcm)	Runoff available (mcm)	Non-committed Runoff available (mcm)	Number of recharge shaft/ tube well/ NB/CD/CP/ percolation tank/ponds
	10.17	13.53	80.45	24.13	recharge shaft/ tube well- 135
					percolation tank- 14
					NB/CD/CP- 225
					Pond/Dug well -68

6.2.d Projected status of Ground Water Resources & Utilization:

After interventions of AR structure the following output comes in respect of stage of development. The details of output are given in Table.25

Table.25 Projected status of ground water resources and its utilisation in Golmuri-Cum-Jugsalai block

Block	Net Ground water availability	Additional recharge from water conservation & AR structure	Total net ground water availability after intervention	Existing Gross Ground water Draft for all uses	Net ground water draft after intervention	Present stage of ground water development	Projected stage of ground water development after interventions
Golmuri -Cum -Jugsalai	ham	ham	ham	Ham	ham	%	%
	1505.1	1017	2522	2218.79	2218.79	147.42	88

6.2.e. Matching of Demand:

Ground Water requirement/demand and its matching/fulfillment pattern to maintain stage of development at 70% of over-exploited block the volume of water required is given in *Table 26*

Table 26 Details of Matching Demand Scenario

Block	Requirement of ground water to maintain stage of development 70% (ham)	Volume of water conserved after adopting roof top rain water harvesting (ham)	Volume of water conserved from construction of Artificial Recharge Structures (ham)	Matching of Demand (ham)
Golmuri-cum-Jugsalai	A	B	C	(B+C)
	1165	340	1017	1357

6.3 Demand side management/Change in cropping pattern:

In the study area of phase III covering East-Singhbhum, Saraikela-Kharsawan and Parts of West Singhbhum district the net dynamic ground water availability in the area is 48667.55 ham and ground water draft for all uses is 9188.63 ham, making stage of ground water development 19% . Existing gross ground water draft for irrigation is 2652 ham which is very less as compare to total net ground water availability in the area. There are ample scope of ground water development and irrigation potential creation without adaptation of advance irrigation practices or change in cropping pattern. However Ground water draft for irrigation may need to be accelerating with extraction of ground water through dug wells for irrigation.

6.4 Management plan for Safe blocks:

Management Strategies:

The following management strategies proposed to be adopted for safe blocks of phase III area.

- Ground water development and irrigation potential creation
- Ground water draft for irrigation may continue with same pace
- No change in Industrial and Domestic ground water draft.
- No change in cropping pattern.

6.4.1 Plan for Ground Water Development:

The net dynamic ground water availability in the area is 48667 ham and ground water draft for all uses is 9188 ham, making stage of ground water development 19% as a whole for the phase III area. Although the ground water development in central part is more, there are ample scope of ground water development in eastern, northern, southern and western part and some part of the central area. The Net ground water availability for future irrigation development and stage of ground water development of safe blocks are given in *Table 27*

Table-27 Net GW availability for future irrigation development and SOD

District	Assessment unit	Net ground water availability for future irrigation development in ham	Stage of ground water development (%)
East-Singhbhum	Bahargora	6230.09	15.37
	Boram	564.37	25.70
	Chakulia	8277.08	5.83
	Dalbhumgarh	1025.50	11.37
	Dumaria	1490.42	7.44
	Ghatsila	1188.92	32.78
	Gurabandha	1029.46	9.62
	Musabani	974.55	39.81
	Potamda	850.94	22.99
	Potka	4132.68	8.10
Saraikela-Kharaswan	Adityapur (Gamhariya)	1247.03	37.40
	Chandil	1159.69	40.85
	Gobindpur	3198.92	9.00
	Ichagarh	1017.77	17.80
	Kharsawan	763.69	22.91
	Kuchai	1424.65	8.24
	Kukru	498.25	20.72
	Nimdih	604.86	25.78
	Saraikela	1136.56	18.18
West Singhbhum	Bandgaon	1512.40	10.90
	Chaibasa	989.25	13.13
	Chakardharpur	1636.06	15.86
	Khuntpani	1007.51	12.96
	Tantnagar	1092.82	11.93

The average stage of ground water development is 19.9%. All 24 blocks, there having stage of development is less than 60% has been considered for further ground water development. District wise balance ground water for future irrigation potential is determined by deducting ground water draft for irrigation and allocation for next 25 years. The ground water available for future irrigation is divided by an average depth of irrigation (Δ), considering of 0.45m for Jharkhand, which ultimately gives irrigation potential (*Table.28*). Considering 70% of future irrigation potential as optimum utilization with 0.45 ha area for dug well and considering 60% for dug well, the numbers of proposed ground water abstraction structures are obtained.

Table 28 Future irrigation potential created and proposed number of dug wells

District	Assessment unit	Net ground water availability and future irrigation development (ham)	future irrigation potential available (ha) considering (Δ)0.45m	70% of future irrigation potential created (ha)	Proposed number of ground water structure (Dug wells)
East-Singhbhum	Bahargora	6230.09	13844	9690	12919
	Boram	564.37	1254	877	1169
	Chakulia	8277.08	18393	12875	17166
	Dalbhumgarh	1025.50	2278	1594	2125
	Dumaria	1490.42	3312	2318	3090
	Ghatsila	1188.92	2642	1849	2465
	Gurabandha	1029.46	2287	1600	2133
	Musabani	974.55	2165	1515	2019
	Potamda	850.94	1890	1323	1764
	Potka	4132.68	9183	6428	8570
Saraikela-Kharsawan	Adityapur (Gamhariya)	1247.03	2771	1939	2585
	Chandil	1159.69	2577	1803	2403
	Gobindpur	3198.92	7108	4975	6633
	Ichagarh	1017.77	2261	1582	2109
	Kharsawan	763.69	1697	1187	1582
	Kuchai	1424.65	3165	2215	2953
	Kukru	498.25	1107	774	1032
	Nimdih	604.86	1344	940	1253
	Saraikela	1136.56	2525	1767	2355
West Singhbhum	Bandgaon	1512.40	3360	2352	3135
	Chaibasa	989.25	2198	1538	2050
	Chakardharpur	1636.06	3635	2544	3391
	Khuntpani	1007.51	2238	1566	2088
	Tantnagar	1092.82	2428	1699	2265
Total		43053.47	95662	66950	89254

Development of ground water for the safe blocks in phase III area requires thorough understanding of the heterogeneity of the formations, e.g degree of weathering, thickness of fracture zones and depth of occurrences of the aquifer. The feasibility of different ground water structures together with yield prospects for the balance ground water resources for future irrigation use is given in (*Table 29*).

Dug wells are feasible ground water structures for the study area. The construction of 89254 additional ground water abstraction structures would bring an additional area of

40164 ha under assured irrigation. This would involve a total expenditure of Rs 423 crore (Table 30)

Table 29 .Proposed feasible GW abstraction structures based on balance potential

GW Balance to be utilized for irrigation (ham)	Type of structure	Depth range (m)	Diameter (m)	Probable yield range (lps)	Command/No on Command area (ham)	No of GW abstraction structure to be constructed
40164	Dug well	12 to 20	3 to 5.5	2 to 5	0.45	89254

Table 30 .Proposed cost estimation for feasible GW abstraction structures

S.No.	Description	No of GW abstraction structure	Unit cost (Rs)	Construction Cost in Crore (Rs)
1	Construction of dug wells fitted with pump set	89254	80000	714
	Total			714
2	Annual maintenance cost	---	5000	44
3	Grand Total			758

7.0 FINDINGS AND RECOMMENDATION:

7.1 Findings: followings are the major findings in the study area

- The area is predominantly underlain by Singhbhum granites with criss-crossing dolerite dykes, metasedimentaries formations.
- The area ranges in altitude from 213 to 945 m and is characterized by widely varying geomorphic features like structural hills, plateau and undulating plains.
- The area is influenced by two major thrusts i.e Singhbhum shear zone and Dalma thrust and different linear structural features.
- Weathered mantle, saprolite zone, shallow and deep fractures have been identified as repositories of ground water in the study area. The weathered mantle which has variable thickness in different litho-units is quite extensive in Chakradharpur granite. In the meta sedimentaries the average thickness of the weathered residuum is 16m while in Singhbhum granite it is 12m. It constitute the shallow aquifers which contain ground water under phreatic condition. The computed hydraulic conductivity for weathered mantle of meta sedimentaries ranges from 1.4 to 2.23×10^{-1} m/day, whereas in case of Singhbhum granite it is 4×10^{-2} m/day. The saprolite zone is persistent throughout the area with the average thickness is around 4m but seldom exceed 8m, with the younger granite showing greater thickness. The ground water occurs under semi-confined conditions. The discharge varies between less than 1 to >3 lps for draw down between 10 and 18 respectively. The hydraulic conductivity ranges from 1.3 to 2.5×10^{-1} m/day with the average storage coefficient being 1×10^{-2} m/day. In hard rock terrain fractures have been identified down to 110 to 140 m in Singhbhum granite and metasediments with the exception at Chakardharpur area, where the fractures have been identified down to 177m. The productive fractures are broadly aligned in NE-SW, NW-SE and E-W directions. The frequency of the fractures decreases with depth. The discharge from fractures varies between 3-8 lps.

- Mode of occurrence of ground water in the weathered mantle, saprolite and fractures zones are different, the total aquifer system is in hydraulic continuity , locally or regionally. Thus, the recharge to the bed rock fractures is from the weathered mantle through the saprolite zone.
- Most of the blocks are irrigation deprived area i.e low ground water developed blocks.

7.2 Recommendations:

- It is recommended that selection of well sites in hard rock terrain should invariably be backed by multidisciplinary investigations involving geophysical, hydrogeological and remote sensing technique.
- Large diameter dug wells are recommended for exploitation of ground water tapping both weathered mantle and saprolite zone. The structure should consist of an open well tapping the entire saturated thickness of weathered mantle with atleast 4m of the saprolite zone. Such wells are suitable for domestic and agricultural purposes.
- Both open and shallow bored wells are recommended in thick quartz of pegmatite veins and dykes for domestic and limited supply of water for agriculture.
- Construction of bore wells tapping bed rock fractures down to 60m depth along with the weathered mantle and saprolite zone are recommended for all geological formations in the area.
- Where weathered mantle and saprolite zone are not pronounced, deep bore wells are recommended down to a depth of 140m in Singhbhum granite and metasediments for restricted pumping.
- The average stage of ground water development is 19%. All 24 blocks, there having stage of development is less than 60% has been considered for further ground water development upto 70%.
- Ground Water Development is required in most irrigation deprived area i.e low ground water developed blocks.
- Schists are devoid of fractures and borewells are unsuccessful in those area. Deep bore wells drilling should be avoid in schist.

Annexure- I**Basic Analysis Result of twenty-six Water Samples collected in Phase III Study area of East Singhbhum, Saraikela-Kharsawan and Parts of West-Singhbhum Districts**

S.n	District	Location	pH	EC	TDS	TH	Ca2+	Mg2+	Na+	K+	CO32-	HCO3-	Cl-	NO3-	SO42-	F-	PO43-	SiO2
1	E. Singhbhum	Bhargora	8.44	400	260	155	42	12.15	22.3	0	8	79.95	70.9	2	2	1.32	0	10
2	E. Singhbhum	Chakulia	7.51	314	204.1	130	38	9.72	15	1.1	0	79.95	35.45	24	18	0.18	0	12
3	E. Singhbhum	Ghatsila	8.17	850	552.5	310	84	24	41	16	0	301.35	95.71	6	0	0.55	0	22
4	E. Singhbhum	Dalbhumgarh	7.46	135	87.75	50	16	2.43	9.4	2.4	0	61.5	7	8	0	0.01	0	12
5	E. Singhbhum	sakshi	7.2	1420	923	490	140	34	88	12	0	375.15	135	66	154	0.48	0	21
6	E. Singhbhum	Potka	8.55	1039	675.4	420	100	41.31	51	1.6	33	354	71	1	35	0.65	0	34
7	E. Singhbhum	Sudernagar	8.41	862	560.3	350	80	38.88	28	3.2	15	123	163	15	34	0.22	0	37
8	E. Singhbhum	Ramgarh	8.52	1052	683.8	500	52	65.61	30	11	42	293	84.98	6	106	0.11	0	33
9	E. Singhbhum	Kalikapur	7.12	1866	1213	540	150	40	155	24	0	430	350	18	0	0.98	0	27
10	E. Singhbhum	Hata/tirin	8.66	1208	785.2	410	140	14.58	75	25	36	307.5	162.45	11	0	0.44	0.22	33
11	Saraikela	Chaoil	8.48	815	529.8	330	90	25.51	44	3.2	36	184.5	124.55	6	0	0.15	0	18
12	Saraikela	Rajnagar	8.22	940	611	400	104	34	33	1.6	0	412	67	33	9	0.33	0	48
13	Saraikela	Saraikela	8.22	937	609.1	400	104	34	41	1.8	0	412	67	28	9	0.25	0	48
14	Saraikela	Keshargaria	8.12	608	395.2	230	52	24	36	2.7	0	172.2	74.45	25	0	0.14	0	18
15	E. Singhbhum	Aoharia	7.74	422	274.3	180	60	7.29	16	12	0	147.6	34.45	17	21	0.11	0	24
16	E. Singhbhum	Lohasoli	8.24	685	445.3	245	50	27.94	60	4.6	0	202.95	84.89	16	28	0.64	0	11
17	E. Singhbhum	Galudih	8.28	477	310.1	205	50	19.44	33	8.65	0	86.1	81.45	36	11	0.78	0	14
18	Saraikela	Jamdih	8.48	585	380.3	220	60	17	44	3.8	24	183	67	17	0	0.22	0	22
19	E. Singhbhum	Pithajudi	8.16	321	208.7	90	20	9.72	33	11	0	135.3	24.81	21	0	0	0	15
20	Saraikela	Kharsawan	8.22	1233	801.5	310	60	38.88	98	35	0	479.7	109.45	22	24	0.47	0	26
21	W. Singhbhum	Chakradharpur	7.83	928	603.2	350	74	40.09	56	1.1	0	356.7	95.55	24	15	0.24	0	40
22	W. Singhbhum	Chaibasa	7.6	1000	650	245	68	18.22	110	44	0	492	95.45	2	0	0.72	0	52
23	W. Singhbhum	Khuntpani	8.22	932	605.8	400	104	34	35	1.8	0	399.75	67	42	9	0.39	0	47
24	W. Singhbhum	Kerekikala	8.23	698	453.7	285	70	26.73	36	18	0	159	99.16	22	48	0.42	0	40
25	W. Singhbhum	Bandgaon	8.21	1482	963.3	375	80	42.52	158	17	0	129.15	390.5	55	44	0.72	0	27
26	W. Singhbhum	Bangaon	8.54	460	299	195	46	19.44	18	10.54	15	159.9	35.45	8.03	14	0.64	0	15

Annexure-II**Result of Trace Metal Concentration in thirty-one Water Samples of Industrial Cluster falling in the study area in East Singhbhum, Saraikela-Kharsawan and Parts of West-Singhbhum Districts**

S. No	Sample ID	Sample Code	Fe	Mn	Al	Ba	Sr	Li	Pb	Zn	Cu	Cr	Co	Ni	Cd	As	Se	Be	Ag	V	Rb
			Trace Metal Concentration in $\mu\text{g L}^{-1}$ (ppb)																		
1	J1	539A/1	1171	532.0	16.1	35.9	283.1	6.7	1.4	218.1	1.0	3.7	0.71	6.1	0.05	4.20	1.15	0.01	0.01	20.2	1.86
2	J2	539A/2	1090	108.2	38.4	33.1	235.3	9.6	3.1	81.3	1.2	1.4	0.47	5.4	0.18	4.54	1.64	0.16	BDL	8.4	1.22
3	J3	539A/3	644	140.5	26.5	29.1	122.7	15.7	0.7	29.9	2.8	3.8	0.40	4.3	0.01	8.69	2.32	0.01	0.04	25.5	0.76
4	J4	539A/4	579	6.4	35.1	21.3	107.7	0.5	1.9	291.0	0.7	2.2	0.26	2.9	0.09	3.60	1.19	BDL	BDL	7.8	1.68
5	J5	539A/5	1058	152.9	12.8	43.6	328.5	25.2	0.6	27.7	0.9	3.5	0.43	5.0	BDL	3.00	0.76	0.02	BDL	7.9	1.15
6	J6	539A/6	955	13.1	74.2	58.8	312.9	5.6	2.0	181.7	1.3	4.1	0.45	5.0	0.06	2.33	0.71	0.02	0.00	7.4	0.65
7	J7	539A/7	746	7.4	39.1	53.5	335.3	3.0	1.7	108.0	3.4	3.9	0.32	4.5	0.05	3.87	2.93	0.00	0.00	9.1	7.63
8	J8	539A/8	670	20.0	297.7	26.6	89.9	4.2	3.1	245.4	5.6	4.2	0.59	9.0	0.11	3.44	1.44	0.04	BDL	10.2	6.41
9	J9	539A/9	1398	6.8	79.1	42.6	364.0	1.6	1.7	61.0	2.7	2.0	0.61	7.1	0.06	4.23	4.14	0.03	BDL	10.6	24.43
10	J10	539A/10	1105	703.9	21.3	13.2	392.5	18.4	3.5	23.1	182.2	1.6	1.69	7.5	0.05	4.10	1.53	0.08	0.08	6.3	3.53
11	J11	539A/11	878	38.3	448.5	71.0	129.9	2.1	1.5	272.7	25.8	2.7	0.68	6.5	0.11	2.63	1.48	0.07	0.01	7.7	8.14
12	J12	539A/12	971	729.2	41.4	59.5	264.8	2.2	0.7	31.0	1.3	2.9	3.29	5.5	0.01	5.14	1.27	0.04	0.01	11.1	0.95
13	J13	539A/13	889	23.2	34.6	27.5	292.7	2.9	6.1	1778.4	1.5	2.9	0.44	7.2	0.88	2.61	2.67	0.04	0.04	6.1	4.18
14	J14	539A/14	2198	74.9	60.5	20.4	876.3	6.9	2.3	108.8	9.7	4.0	1.31	13.5	0.16	3.85	1.30	0.04	BDL	9.3	1.05
15	J15	539A/15	1187	10.2	17.5	42.0	305.5	3.8	0.1	19.9	1.0	0.8	0.50	6.7	0.00	2.28	1.10	0.02	BDL	5.3	2.21
16	J16	539A/16	3010	1135.7	38.7	219.1	894.6	3.3	7.5	82.7	14.3	2.3	1.96	13.3	0.32	7.19	1.60	0.22	0.27	12.7	0.81
17	AG 1	540A/1	2337	414.5	20.6	156.6	467.8	8.8	2.0	107.1	6.0	0.9	8.56	32.7	0.06	6.15	1.09	0.08	0.04	13.1	0.55
18	AG 2	540A/2	1340	141.0	112.7	45.2	409.4	6.4	4.6	173.4	4.5	2.7	0.94	8.1	0.27	6.23	2.12	0.07	0.00	17.1	2.57
19	AG 3	540A/3	898	28.9	16.6	115.3	301.7	15.5	0.1	42.4	1.4	2.3	1.74	5.8	0.01	3.53	1.02	0.03	0.00	11.7	0.84
20	AG 4	540A/4	2873	295.9	37.8	95.2	1067	9.2	1.9	210.4	1.7	2.8	1.94	17.8	0.16	6.90	1.73	0.12	BDL	18.0	2.02
21	AG 5	540A/5	6560	531.7	38.0	292.3	1248	23.0	20.	165.7	9.3	5.9	4.31	29.2	0.26	10.43	1.69	0.09	0.07	17.0	0.96
22	AG 6	540A/6	932	38.0	28.3	24.7	239.5	2.9	1.5	228.9	1.5	1.7	0.47	5.2	0.03	2.98	0.82	0.03	BDL	8.2	2.04
23	AG 7	540A/7	1326	182.0	8.9	98.3	347.5	16.0	6.2	1319.3	0.7	1.6	1.49	4.5	0.01	4.40	1.43	0.02	0.01	10.7	0.56
24	AG 8	540A/8	2639	49.5	16.1	29.6	432.5	6.5	1.8	1600.7	8.0	2.5	0.50	5.7	0.09	4.40	1.60	0.04	0.02	6.9	9.31
25	AG 9	540A/9	1981	18.0	32.8	39.4	512.7	14.2	2.4	129.2	15.9	2.8	0.60	8.3	0.10	5.65	1.24	0.04	BDL	15.6	1.49
26	AG 10	540A/10	679	118.0	37.6	98.9	156.5	2.3	0.1	31.4	1.7	2.6	0.40	4.7	0.03	3.69	1.20	0.03	0.01	8.7	7.29
27	AG 11	540A/11	2648	78.9	43.4	37.8	720.2	13.2	2.6	78.2	4.2	2.5	1.47	11.5	0.17	5.18	20.15	0.17	0.19	24.9	0.61
28	AG 12	540A/12	1718	11.6	34.0	5.9	157.5	2.5	1.3	44.5	3.0	3.7	0.28	3.8	0.03	4.68	1.19	0.01	0.16	46.0	1.00
29	AG 13	540A/13	3022	481.4	1813.	62.4	114.1	7.9	4.2	213.3	5.3	6.6	5.63	9.4	0.07	5.80	1.22	0.20	0.10	21.8	4.30
30	AG 14	540A/14	2904	815.3	345.6	36.9	912.1	5.3	2.3	929.6	3.6	3.8	6.50	24.4	0.22	6.34	4.70	0.12	0.10	15.1	1.48
31	AG 15	540A/15	937	93.0	54.8	42.8	294.7	9.3	2.1	111.9	3.8	1.9	0.71	5.5	0.04	5.03	1.59	0.05	0.05	16.1	0.77

Samples are analyzed by Perkin Elmer ICP-MS (Model: ELAN DRGe)

Annexure-III

Exploration Details of East Singhbhum, Saraikela-Kharsawan and Parts of West-Singhbhum Districts

Location(Name of the site)	Block	District	Formation (Geology)	Long	Lat	Depth Drilled (mbgl)	Length of Casing pipe/ Depth const.	Type	Aquifer Type & depth range (mbgl)	Granular/Zone/ fracture Tapped	Discharge (lps)	Static Water level (mbgl)	Draw down	Specific Capacity	Transmissivity	Storage
Hesel	Potka	E. Singhbhum	Singbhum Granite	86.15278	22.45833	204	3.86	EW	W1:-39 F1:-59, F2:-83, F3:-105, F4:-131, F5:-140	39,59,83,105,131,140	W1:-0.41 F1:-0.45, F2:-0.4, F3:-1, F4:-2.1, F5:-5.9					
Kudada	Jamshedpur	E. Singhbhum	Metasediments	86.20639	22.75111	145.41	15.6	EW	F1:-105, F2:-117, F3:-137	105,117,137	F1:-0.41, F2:-0.6, F3:-11.5					
Kalapathar	Chakulia	E.Singhbhum	Metasediments	86.77306	22.41111											
Ulda	Potka	E. Singhbhum	Metasediments	86.17222	22.65139	137.97	16.25	EW	F1:-40.91-41.91, F2:-65.77-66.77, F3:-129-130	40.91-41.91, 65.77-66.77, 129-130	F1:-0.75, F2:-1.58, F3:-9.16					
AMD Campus, Jamshedpur	Jamshedpur	E. Singhbhum	Metasediments	84.81986	23.53667	196	41.7	EW	F1:-80-81, F2:-133.20-134, F3:-152.50-153, F4:-160-161	80-81,133.20-134,152.50-153,160-161	F1:-0.45, F2:-1.35, F3:-1.6, F4:-16	12.4	8.2	109.0Lpm/m	101m ² /day	7.44 x10 ³
Haldipokhar	Potka	E. Singhbhum	Singbhum Granite	86.14167	22.59861	200	19.4	EW	F1:-159.50-160.80	159.50-160.80	F1:-megre					
Potamda Block Office	Potamda	E. Singhbhum	Metavolcanics	86.38222	22.90333	200	13.5	Ew	F1:-58.75-60.25	58.75-60.25	F1:-megre	6.6				
Manusumaria	Baharagora	E.Singhbhum	Tertiary	86.77778	22.36389	121	-	EW	G.Z.1:-32-33, G.Z.2: 43-44, G.Z.3: 50-51, G.Z.4:71-72	32-33,43-44,50-51,71-72	G.Z.1:-megre, G.Z.2: 0.45, G.Z.3: megre, G.Z.4:0.45	12.3	10.98	5.19 Lpm/m	209 m ² /day	1.30 X10 ⁻⁴
Baharagora	Baharagora	E.Singhbhum	Metasediments	86.73611	22.26944	99.26	-	EW	W1:- 20-21 F1:-61-62	20-21,61-62	W1:-0.45 F1:-0.45	3.6	14.44	5.4 Lpm/m	570.8 m ² /day	
Ghoralinga	Saraikela	Saraikela	Metasediments	85.96833	22.66944	202	16	EW			Dry					
Dugni	Adityapur	Saraikela	Metasediments	85.97361	22.75833	191.1	24.5	EW	F1:-107	107	F1-0.75					
Block office campus (kuchai)	Kuchai	Saraikela	Singbhum Granite	85.74861	22.84167	200	30.45	EW	W1:- 46-47, F1:-65-65.50	65-65.50	W1:-3.4, F1:-.45	7.56	20.73	9.1 Lpm/m	14.0 m ² /day	1.24 x10 ³
Govt.High school (Chouka)	Chandil	Saraikela	Singbhum Granite	85.96356	22.95769	200	24.4	Ew	W1:- 25.20-26.30 W2:-39.80-40.20	25.20-26.30, 39.80-40.20	W1:-0.45 W2:-0.45					

Murkum Panchayat Sachivalaya	Saraikela	Saraikela	Singbhum Granite	85.93333	22.70056	200	18.2	EW	W1:-37-38 F1:-61.50-62, F2:-132.20-133.50, F3:-169.50-170.50	37-38,61.50-62,132.20-133.50,169.50-170.50	W1:-0.8 F1:-megre, F2:-0.8, F3:-0.45	-				
Rola (Kartik Oraon Girls High school)	Rajnagar	Saraikela	Singbhum Granite	85.96844	22.61972	200	20.3	EW	W1:-30-31.	30-31	W1:-megre	3.65				
Purnadih (Utkramit M.School)	Saraikela	Saraikela	Metasediments	85.93306	23.06611	200	21	EW	F1:-42.8-43.8	42.8-43.8	F1:-0.45	3.3				
Chandil Swaranrekha Dam Campus EW	Chandil	Saraikela	Singbhum Granite	86.01562	22.9529	200	10	EW	W1:-36-37	36-37	W1:-megre					
Chaulibasa (Govt. Middle school) EW	Chandil	Saraikela	Singbhum Granite	85.95182	22.95516	135.70	21	EW	W1:-33-34, F1:-42-43	33-34, 42-43	W1:-1.2, F1:-3.4	6.2				
Ichagarh(Birsa High School) EW	Ichagarh	Saraikela	Singbhum Granite	85.85002	22.97836	150.00		EW			Dry					
Basahatu (Utkramit M.School) EW	Ichagarh	Saraikela	Singbhum Granite	85.83403	22.99868	200.00	25.5	EW	W1:-39-40, F1:-160-161	39-40, 160-161	W1:-3.4, F1:-2.2	5.6				
Basic Seed Multiplication & Training Centre EW	Kharsawan	Saraikela-Kharsawan	Singbhum Granite	85.825	22.781	181.40	13.5	EW	W1:-28.90-31.90, F1:-65.50-66.50, F2:-71.60-72.60, F3:-77.70-78.70	28.90-31.90, 65.50-66.50, 71.60-72.60, 77.70-78.70	W1:-0.45, F1:-1.2, F2:-1 F3:-2.2	8.4				
PWD Guest house	Chakardharpur	W. Singhbhum	Chakardharpur Granite	85.61667	22.68333	209.42	11.5	EW	F1:-176.94	176.94	F1:-4.5					
Bandgaon Block office campus	Bandgaon	W. Singhbhum	Singbhum Granite	85.32647	22.85333	200	14.6	Ew			Dry					
Matamkaber (Govt. Middle School)	Bandgaon	West Singhbhum	Singbhum Granite	85.573	22.708	196.60	25.5	EW	W1:-28.90-29.90, F1:-181.4-182	28.90-29.90,181.4-182	W1:-megre, F1:-0.45	9.6				
Upgraded high school,	Tantnagar	West Singhbhum	Metasediments	85.942	22.451	153.00		EW			1.8					
Purnia Govt. School EW	Khuntpani	West Singhbhum	Metasediments	85.761	22.668	153.80	24	EW	W1:-28.90-29.90, F1:-59.40-60.40, F2:-102.10-103.10	28.90-29.90, 59.40-60.40, 102.10-103.10	W1:-2.2, F1:-0.45, F2:-1.8	6.1				